

Apollo RTCC MFD

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1 Introduction

The Apollo RTCC MFD provides the necessary calculation tools to fly complete Apollo missions with Project Apollo - NASSP 8.0. As much as possible it tries to replicate the same calculations, inputs and display as were used by the actual flight controllers during Apollo. Originally started to calculate the Apollo 7 rendezvous maneuvers, the MFD has expanded to include many more features which during the Apollo program were provided by Mission Control (MCC) and the Real-Time Computer Complex (RTCC).

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2 Main Menu

The main menu is dividing the MFD in the following categories:

TAR: Targeting menu. Contains the various maneuver computation pages.

PAD: Pre-Advisory Data. Shows the PADs that the Apollo crews received during a mission.

UTI: Utility. All additional calculation pages that are not for specific maneuvers.

INI: MPT Initialization. Used to update weights, areas and vehicle configuration for the MPT. Also used for non-MPT trajectory propagation.

PLN: Mission Plan Table. A central feature of the maneuver planning during a mission. Currently optional.

CFG: Configuration. Various settings for the MFD.

UPL: Uplink Page. All uplinks to the AGCs and LVDC can be found here.

MCC: MCC Displays. Shows the "TV Guide", a list of displays that were available in the MOCR.

3 Targeting

The targeting menu consists of the many maneuver calculation pages:

REN: Rendezvous menu. Contains the calculations for rendezvous maneuvers.

ORB: Orbit Adjustment. Contains the inputs and display for the General Purpose Maneuver processor.

TLI: TLI Planning. Currently under construction.

MCC: Midcourse Correction. Contains the inputs and displays for the Translunar Midcourse Correction Processor.

LOI: Lunar Orbit. Contains the inputs and displays for the Lunar Orbit Insertion processor.

ENT: Entry. Contains the inputs and displays for the Return-to-Earth processor (RTEP).

DEO: Deorbit targeting. Contains the inputs and displays for the Retrofire Planning.

DES: DOI Targeting. Contains the inputs and display for the Lunar Descent Planning Processor (LDPP).

LIF: Lunar Liftoff. Contains the inputs and display for the Lunar Launch Window Processor (LLWP).

ASC: Lunar Ascent. Contains the inputs and display for the Lunar Ascent Integrator (LAI).

ABO: Descent abort. Contains the inputs and display for the Powered Descent Abort Program (PDAP).

3.1 Rendezvous

This MFD page contains the three main calculation tools for rendezvous maneuvers, as well as a separate section for the Skylab rendezvous profile:

LAM: Lambert targeting. Contains the inputs and display for the Two-Impulse (TI) processor.

CDH: CDH/NSR maneuver. Contains the inputs and display for the Coelliptic Rendezvous processor (SPQ).

DKI: Docking initiate. Contains the inputs and display for the Docking Initiation Processor (DKI)

SKY: Skylab rendezvous. Replicates the rendezvous programs of the onboard software in the AGC of the Skylab missions.

3.1.1 Two-Impulse Processor (TI)

The purpose of the Two-Impulse section of the RTCC is to provide the flight controller with a set of two maneuvers to achieve desired orbit conditions of the maneuvering vehicle. The section works in terms of chaser and target vehicle, and in terms of a transfer initiation time and transfer completion time. The first maneuver computed is a chaser transfer initiation maneuver needed to intercept a point in space which is defined in terms of the target vehicle. This intercept point is defined using phase angle and height difference offsets from the target vehicle at the transfer completion time. The second maneuver of the set, which is done at this intercept point, is a coelliptic maneuver. For example, if the desired offsets are both zero at the transfer completion time, the transfer initiation maneuver will cause the chaser to intercept the target at transfer completion time, and the transfer completion maneuver would be a velocity matching maneuver.

The Two-Impulse section has two major areas to accomplish its purpose: (1) the regular Two-Impulse or Multiple Solution option, and (2) a Corrective Combination or Digital Option.

The regular Two-Impulse request generates a maximum of thirteen solutions that may be displayed on the Multiple Solution Display. If more than thirteen solutions are computed, the thirteen that require the least fuel expenditure are displayed. The flight controller may fix the time of the first maneuver and vary the time of the second maneuver through a time increment; he may hold the second maneuver time fixed and vary the first maneuver time; or he may compute only one solution at the input times. An option also exists to compute the time of the first maneuver in terms of an elevation angle to the target. The time of the second maneuver may also be computed in terms of a target vehicle central angle of travel from transfer initiation time to transfer completion time. These two options are normally used for the terminal phase maneuvers, i.e., zero offsets.

The Corrective Combination area of the Two-Impulse section is normally used to obtain satisfactory coelliptic conditions prior to terminal phase. This may be used when launch dispersion or improper maneuver execution may cause other rendezvous techniques some problems. There are two methods of obtaining the optimum corrective combination solution.

1. The first method requires a range of height differences to be input. A range of times for the transfer completion maneuver are also input, and the phase angle for each time is computed. For each height difference the range of solutions are computed, and the impulsive solution at each height difference that requires the least amount of fuel expenditure is displayed.
2. The second method involves a series of solutions that also requires a range of height differences at a transfer completion time. For each height difference requested, a range of phase angles at transfer completion time is used, and the optimum solution for each height is displayed. This range of phase angles for each height is computed from a time limit on each side of terminal phase initiation, with an input increment determining the number of solutions attempted. A terminal phase slip time is computed for each solution displayed.

Another display that is driven by this section is the Single Solution Display. The flight

controller may view any one solution from either the multiple or corrective combination table. This display contains components in external DV coordinates, day light-darkness information, and pitch and yaw angles so that the window is along a. given line of sight, and contains range, range rate and azimuth approach data at requested times.

Two Impulse Computation (K30)

Inputs:

VEH: Choose the chaser vehicle, CSM or LEM.

IV: Maneuver fixing mode. 0 = Time of both maneuvers fixed, 1 = time of first maneuver fixed, 2 = time of second maneuver fixed

CVT: Chaser vector time. With the MPT inactive this option can be used to select the chaser vehicle.

TVT: Target vector time. With the MPT inactive this option can be used to select the target vehicle.

T1: Transfer initiation time. Set to a negative value for the processor to find T1 as the time at which the elevation angle from the chaser to the target equals the input target elevation angle (P51 button).

T2: Transfer completion time. Set to a negative value for the processor to find T2 as the time after the first maneuver required for the target vehicle to traverse through an angle equal to the input target transfer angle (P51 button).

INC: Time increment between the variable maneuver times; used if one of the maneuver times is not fixed.

RAN: Time range over which the variable time maneuver is to be computed; used if one of the maneuver times is not fixed.

P51: Offsets and elevation angle for two-impulse solution. Special logic is used if all inputs here are non-zero. In this case the assumption is made that the maneuvers are a NCC/NSR combination with the eventual goal of performing a terminal phase intercept (TPI) maneuver. The multiple solution display then shows the predicted TPI time instead of pitch and yaw angles of the second maneuver. The parameters are the delta height in NM and phase angle in degrees, as well as the target elevation angle at the first maneuver point (deg), and the target angle of travel from the first to the second maneuver (deg).

Display:

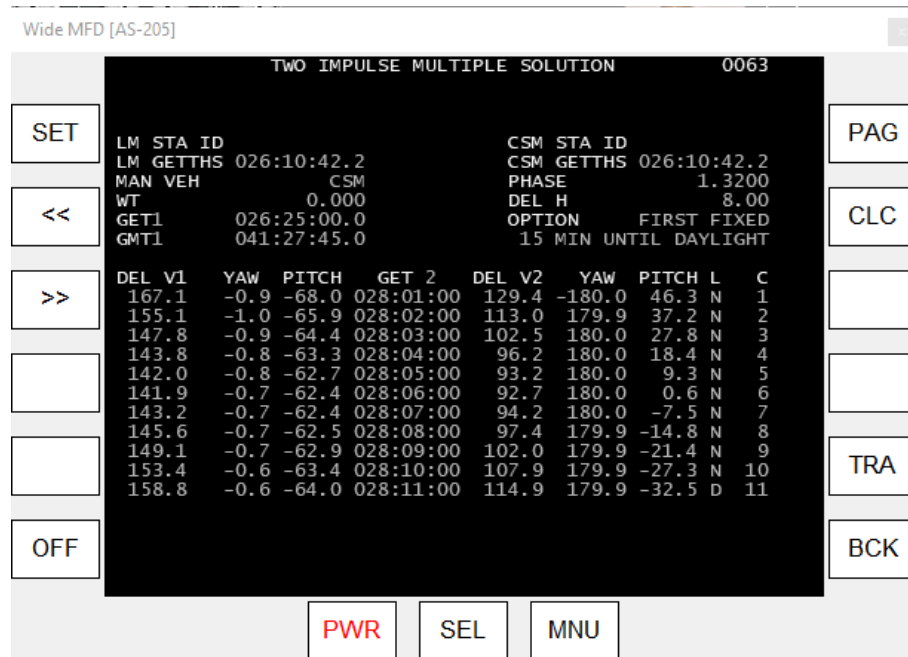


Figure 1: Multiple Solution Display

LM STA ID: standard seven-character ID code indicating the LM anchor vector radar station at the time the display quantities were generated

CSM STA ID: same as above except for CSM

LM GETTHS: time tag of the vector from the current ephemeris table to be used in computing the maneuver, hr:min:sec

CSM GETTHS: same as above except for CSM

MAN VEH: maneuvering vehicle

PHASE: desired phase angle at the second maneuver, deg

WT: central angle of travel between the first and second maneuver, deg

DEL H: maneuvering vehicle

GET1: ground elapsed time of the fixed maneuver, hr:min:sec

OPTION: ID indicating whether the first, second, or both maneuver times are fixed

GMT1: Greenwich mean time of the first maneuver, hr:min:sec

DAY/NIGHT INFO: time until daylight or darkness for the fixed time maneuver, min

DEL V1: impulsive DV for the first maneuver, fps

YAW: yaw angle of the first maneuver, deg

PITCH: pitch angle above local horizontal of the first maneuver, deg

GET2: ground elapsed time of the variable maneuver, hr:min:sec

DEL V2: impulsive DV for the second maneuver, fps

YAW: yaw angle of the second maneuver, deg

PITCH: pitch angle above local horizontal of the second maneuver, deg

TTPI: With the special display logic explained above, the time of TPI is displayed instead of yaw and pitch of the second maneuver, hr:min:sec

Corrective Combination (K32)

Inputs:

VEH: Choose the chaser vehicle, CSM or LEM

REQ: Request type. 0 = second maneuver time incremented between the minimum and maximum times, 1 = second maneuver time is fixed

CVT: Chaser vector time. With the MPT inactive this option can be used to select the chaser vehicle

TVT: Target vector time. With the MPT inactive this option can be used to select the target vehicle

NCC: The time of the first maneuver (GET in hrs:min:sec)

DHMIN: Minimum target height offset at the second maneuver point (NM)

DHMAX: Maximum target height offset at the second maneuver point (NM)

DHINC: Height increment between minimum and maximum height offset (NM)

T2MIN: Minimum (mode 0) or actual (mode 1) time of the second maneuver (GET in hrs:min:sec)

T2MAX: Maximum time of the second maneuver for mode 0 (GET in hrs:min:sec)

DT: Time increment for incrementing the time of the second maneuver in mode 0 or for slipping TPI time in mode 1 (min)

SLIP: The amount by which TPI time is allowed to slip on either side of the nominal time for mode 1 (min)

NOTE: If the mode designation is (1), thus fixing the second maneuver time at the input minimum time, TPI time is allowed to slip over the input range; whereas, if the mode designation is, (0), the time of the second maneuver is varied over its input range. For either mode, the time of the first maneuver is fixed, and the solution which requires the minimum DV for each height offset requested is displayed as the solution for that height.

NOTE: The total DV (DVT) calculation for this display uses a curve fit to estimate TPI and TPF DVs. In this curve fit the terminal phase travel angle (WT) entered on the P51 MED is used. This is the WT on the multiple solution input page.

P52: Two-Impulse Corrective Combination Nominals. Input of the offsets is required for the corrective combination calculation. The inputs are the nominal time of NSR (GET in hrs:min:sec) and the nominal target height and phase angle offsets of NSR (NM and deg, respectively)

Display:

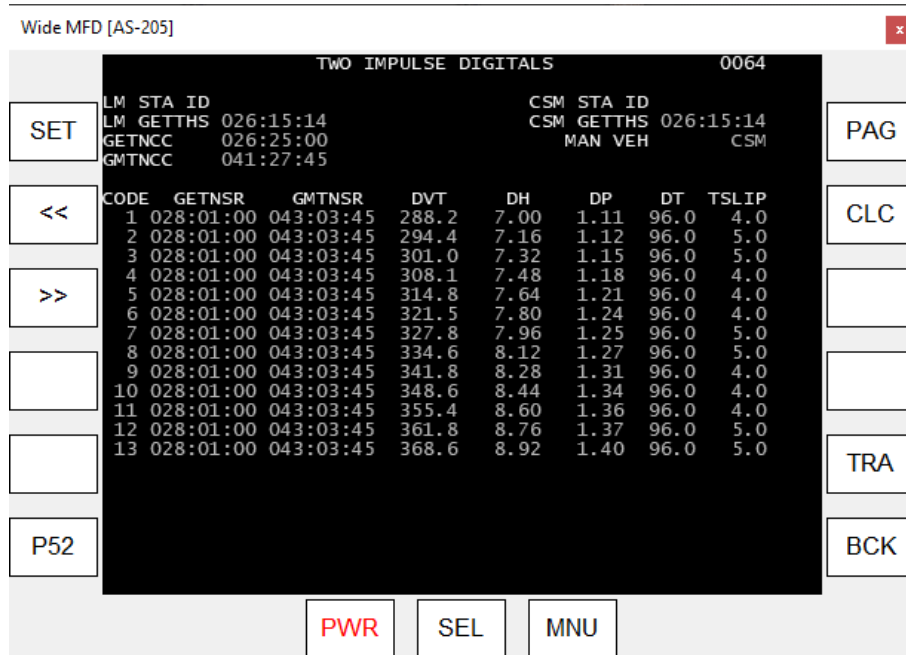


Figure 2: Corrective Combination Display

CSM STA ID: Current CSM STA ID in MPT

LM STA ID: Current LM STA ID in MPT

LM GETTHS: time tag of the vector from the current ephemeris table to be used in computing the maneuver, hr:min:sec

CSM GETTHS: same as above except for CSM

MAN VEH: Maneuver vehicle code

GETNCC: Impulsive elapsed time of first maneuver of all solutions, hr:min:sec

GMTNCC: Impulsive GMT of same, hr:min:sec

CODE: Numeric code specifying the plan number of each solution in the table

GETNSR: Impulsive elapsed time of the second maneuver of the plane, hr:min:sec

GMTNSR: Impulsive GMT of same, hr:min:sec

DVT: Sum of the impulsive DVs for both maneuvers plus an approximation of the DV required for terminal phase

DH: Height offset after the second maneuver, NM

DP: Phase angle after the second maneuver, deg

DT: Time interval between first and second maneuvers, min

TSLIP: Amount of time terminal phase initiation will differ from the nominal if this solution is used, min

Single Solution (K31)

Inputs:

TAB: The two-impulse table indicator, either multiple solution or corrective combination

SOL: Solution number from that table (1 to 13)

QUAD: The number of RCS quads to be used in computing the burn times for the required DV components

LOS: The line-of-sight mode (i.e. whether the component DVs are in a body axes coordinate system defined by the astronauts line-of-sight aligned to the horizon or to the target vehicle)

PIT: The pitch angle (i.e. the angle between astronauts line-of-sight and the X body axis-deg)

DT: The time step between the relative motion data displayed (sec)

Display:

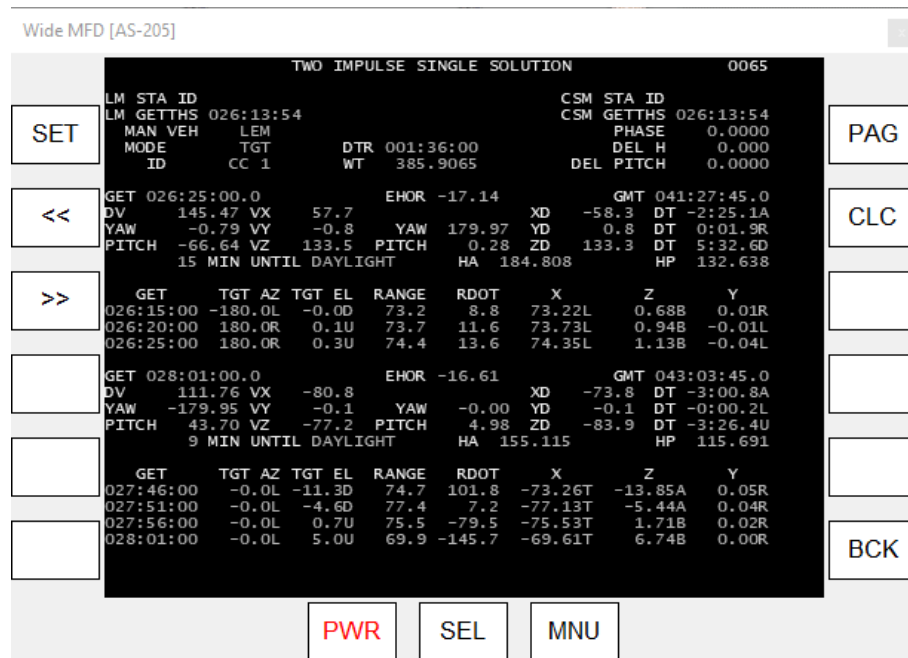


Figure 3: Single Solution Display

LM STA ID: standard seven-character ID code indicating the LM anchor vector radar station at the time the display quantities were generated

CSM STA ID: same as above except for CSM

LM GETTHS: time tag of the vector from the current ephemeris table to be used in computing the maneuver, hr:min:sec

CSM GETTHS: same as above except for CSM

MAN VEH: Maneuver vehicle code

PHASE: actual phase lag at the second maneuver, deg

MODE: ID (either TGT or HOR) indicating the body-referenced coordinate system selected via input for the (XD, YD, ZD) DV definition defined below)

DTR: time between the two maneuvers, hr:min:sec

DEL H: actual differential altitude at the second maneuver time, NM

ID: code identifying the row number of the corresponding solution in the two-impulse multiple solution table or the two-impulse digital display

WT: central angle of travel of the target vehicle between maneuvers, deg

DEL PITCH: angle between the astronaut's line of sight (LOS) and the X-axis of the spacecraft (+ represents a counterclockwise rotation about the Y axis)

GET: ground elapsed time of the impulsive maneuver time, hr:min:sec

EHOR: angle between the astronaut's LOS to the horizon and the local horizontal plane, deg

GMT: Greenwich mean time of the impulsive maneuver time, hr:min:sec

DV: impulsive incremental velocity required by the maneuver, fps

YAW (on the left of display): yaw attitude relative to the local vertical/local horizontal (LVLH) reference frame at main engine ignition required to place the thrust vector along the DV vector (+ to the right), deg

PITCH (on the left of display): pitch attitude relative to the LVLH reference frame at main engine ignition required to place the thrust vector along the DV vector (+ up), deg

VX, VY, VZ: components of the incremental velocity along the X-, Y-, Z-axis, respectively, of either the target-defined or horizontal-defined body axis system, fps

YAW (in center of display): yaw angle between the X-axis of the (in center of display) spacecraft and the local horizontal plane whenever the astronaut's LOS is to the target or the horizon, deg

PITCH (in center of display): pitch angle between X-axis of spacecraft and local horizontal plane whenever astronaut's LOS is to the target or horizon, deg

XD, YD, ZD: components of the incremental velocity along the X-, Y-, and Z-axis, respectively, of either the target-defined or horizontal-defined body axis system, fps

DT (associated with XD): burn time required to impart the velocity increment XD along the X-axis of the body axis system, by firing the fore or aft firing RCS thrusters, min:sec

DT (associated with YD): burn time required to impart the velocity increment YD along the Y-axis of the body axis system, by firing the left or right firing RCS thrusters, min:sec

DT (associated with ZD): burn time required to impart the velocity increment ZD along the Z-axis of the body axis system, by firing the up or down firing RCS thrusters, min:sec

DAY/NIGHT INFO: time until daylight or darkness for the fixed time maneuver, min

HA: apogee height after the maneuver, NM

HP: perigee height after the maneuver, NM

GET: ground elapsed time corresponding to the particular row of the approach data displayed, hr:min:sec

TGT AZ: azimuth of the target vehicle in the chaser vehicle's local vertical/local horizontal coordinate system, deg

TGT EL: elevation of the target vehicle in the chaser vehicle's local vertical/local horizontal coordinate system, deg

RANGE: range from the chaser to the target, NM

R DOT: range rate between chaser and the target, fps

X, Y, Z: position coordinates of the chaser vehicle in a target-centered curvilinear coordinate system, NM The letters L or T, R or L, and U or D are to be printed out with the position coordinates X, Y, and Z, respectively, to indicate whether the target vehicle is leading or trailing, to the right or to the left of, and above or below, the chaser vehicle

3.1.2 Coelliptic Maneuver Processor (SPQ)

Explanation

Coelliptic orbits are two orbits that are coplanar (identical inclination and longitude of the ascending node) and confocal (identical eccentricity and argument of periapsis). To achieve such an orbit relative to a target vessel this MFD can calculate a maneuver based on Program 33 of the AGC and the maneuver to initiate the coelliptic sequence, based on Program 32 of the AGC.

Buttons

INI: Go to SPQ initialization page.

VEH: Choose which of the two vehicles is the chaser and which is the target (MPT mode only).

CHA: Threshold time for the state vector of the chaser vehicle (MPT mode only).

TGT: Threshold time for the state vector of the target vehicle (MPT mode only).

MOD: Calculation mode, CSI or CDH maneuver.

TIM: Switches between fixed GET and finding the delta height of the maneuver or fixed delta height and finding the time of ignition.

TIG: The time of the maneuver in GET. If the option is used to find the CDH time based on delta height this is an initial guess.

CLC: Calculate the burn solution.

BCK: Go back to the main menu.

Init page buttons

DH: Delta height at the CDH maneuver.

E: Desired elevation angle at TPI.

TPI: Desired TPI time.

BCK: Go back to SPQ calculation page.

3.1.3 Docking Initiation Processor (DKI)

Explanation

The basic function of the DKI is to compute impulsive maneuvers; the result of these maneuver is the rendezvous of the CSM and LM spacecraft. The DKI attains a coelliptic orbit by doing three maneuvers: (1) phase, (2) height, (3) a coelliptic maneuver that puts the chaser in a coelliptic orbit with the target. From this orbit a terminal phase maneuver (TPI) and a terminal velocity match maneuver (TPF) may be performed to achieve the actual rendezvous. The Skylab rendezvous sequences with an additional height maneuver is also supported.

The sequence of maneuvers are defined by so called maneuver lines. This terminology stems from the Gemini program. The first apogee after orbital insertion was assigned the number 1.0, the apogee one orbit later was maneuver line 2.0 and so forth. The Gemini 10 rendezvous for example had this maneuver schedule:

1. N=1.0: First apogee
2. N=1.5: Height maneuver (NH)
3. N=2.0: Phase maneuver (NC1)
4. N=3.0: Coelliptic maneuver (NSR)

This results in the actual rendezvous (TPF) happening at roughly the fourth orbit after launch, in the Gemini terminology this was called M=4.

Terminology:

M	The number of spacecraft apogee nearest the rendezvous point (M=4 is a fourth-apogee rendezvous)
N	The in-orbit maneuver line counter. Apogee point is N=X, perigee point is N=X.5, common nodal (plane change) points are N= X.25 or X.75
NH	Apogee height adjustment maneuver performed at perigee end of maneuver line. NH = 1.5 denotes height adjustment near first spacecraft perigee
NC1	Phase adjustment maneuver performed at apogee end of maneuver line to change the catch-up rate. NC1 = 2 denotes phase adjustment near second spacecraft apogee.

TIG and TPI Definition

On this page the time of ignition and time of the TPI maneuver are defined.

VEH: Choose the active vehicle for the rendezvous. Options are CSM or LM. The MFD automatically detects if the current vehicle is a CSM or not.

PRO: For the Skylab rendezvous profile with an additional height adjustment maneuver this option can be set, otherwise the Regular DKI option should be used.

MAN: Options for the initial maneuver line. The line can be defined for an input time, chaser apoapsis or target apoapsis. Input time is useful if the time of ignition or number

of orbits to time of ignition is given. Chaser apoapsis is the typical option for ground-up Earth orbit rendezvous. Should the target be in an elliptical orbit the target apoapsis option is useful to keep the maneuver line near the line-of-apsides of the target, which prevents large vertical Delta V components for the coelliptic maneuver.

ML: Value of the initial maneuver line. This can be chosen arbitrarily, but usually an X.0 value is assigned to a chaser apoapsis. Another example is a CSM rescue maneuver in lunar orbit, which is supposed to happen one orbit after DOI. Then the time of DOI is used for the initial maneuver line, the value entered with this button should be 1.0 and then first maneuver should be scheduled at the 2.0 maneuver line point

PHA: Flag to describe the initial phase angle. Normally set to 0 to indicate a phase angle of -180 to 180 DEG. In this case the chaser needs to catch up if it is behind the target by e.g. 10 DEG. If the chaser is ahead of the target by 10 DEG it needs to slow down, which means a higher orbit. If the flag is set to -1 in the first case the phase angle is interpreted as -350 DEG instead of 10 DEG and the chaser will slow down, go into a higher orbit, to achieve rendezvous. This is mainly useful for rendezvous sequences that take many orbits.

TPD: Definition of the terminal phase. There are three options for both TPI and TPF, so in total six combinations. The three options are input time, time from sunrise and time from sunset. Usually the time of TPI is known from an external calculation or the time from sunrise option with a negative time is used, to set the TPI maneuver in darkness at a fixed time before sunrise.

TPV: Input value associated with the terminal phase definition. Either a GET or a time in minutes from sunset or sunrise.

Init Parameters

DH1: The Delta Height at the NCC maneuver point. Only required for Skylab rendezvous profile. Usually 20 NM

DH2: The Delta Height at the NSR maneuver point. A positive value indicates the chaser vehicle being below the target. Usually 10 to 20 NM.

E: Elevation angle at TPI. Typical value is 26.6 DEG for a rendezvous from below. For rendezvous from above, for example if the CSM needs to rescue the LM in lunar orbit, the value 208.3 DEG is usually used.

WT: Transfer angle from TPI to TPF, typically between 130 and 140 degrees.

MIN: Minimum allowable periapsis altitude in the rendezvous plan.

DKI Inputs

NC1: Choose the maneuver line number of the NC1 maneuver. Set to a negative value if no phasing maneuver is in the maneuver plan.

NH1: Choose the maneuver line number of the NH maneuver. Set to a negative value if no height maneuver is in the maneuver plan.

NSR: Choose the maneuver line number of the NSR maneuver. If the rendezvous profile is the Skylab profile then the NCC maneuver line number is selected here and NSR happens at a specific time after NCC.

MI: The maneuver line number for the rendezvous point (TPF). This is an integer number and only has to be a rough estimate, typically this will be set so that rendezvous occurs

one orbit after NSR.

IDM: Number of additional rendezvous plans that are being generated. For this the NSR and subsequent maneuvers are set to occur one orbit later and the DKI calculation is run again for the updated plan.

MNH: Location of the NH maneuver in the additional plans select with the IDM button. The NH maneuver either stays at its original maneuver line, as specified on this MFD page, or it is being moved one orbit for every additional plan, in effect keeping its place in the plan relative to NSR.

NPC: Location of the plane change maneuver in the rendezvous plan. Not available yet.

Example: Apollo 10 PDI Abort

A simple example for a DKI targeted maneuver is the Apollo 10 PDI Abort. Similar in concept to the PDI+12 maneuvers of the lunar landing missions, Apollo 10 would have done an abort maneuver at perilune after DOI to return to the CSM one orbit earlier than in the normal rendezvous plan. The sequence of maneuvers would be the abort initiation 0.5 revolutions after DOI, targeted as a phasing (NC1) maneuver. CSI or NH would follow another half revolution later, and again the CDH or NSR maneuver would be half an orbit after that. If we assign 1.0 as the maneuver line to the DOI maneuver then we would have $NC1 = 1.5$, $NH = 2.0$, $NSR = 2.5$, $M = 3$. The maneuver line definition is using the time option with the DOI TIG as input.

3.2 General Purpose Maneuver (GPM)

3.2.1 Explanation

The following explanation was taken from IBM RTCC Apollo Programming Systems, Missions Systems, General, Volume II (NTRS ID 19730062603):

The function of the General Purpose Maneuver Processor is to provide the flight controller with two main capabilities:

1. To determine the effect that a specified incremental velocity applied at a given maneuver point (along a given pitch and yaw) will have on the orbit.
2. To determine the maneuver required to obtain a specified orbit or orbital change.

The first capability is more commonly known as a flight controller special-maneuver request and has six options for the maneuver point:

1. An equatorial (nodal) crossing
2. A specified longitude
3. A specified time
4. A specified height
5. An apogee crossing
6. A perigee crossing

The second capability is divided into eight types with various maneuver points:

1. A plane change at a certain equatorial crossing, longitude, time, or height.
2. A circularization maneuver at a longitude or height.
3. A maneuver at perigee to adjust apogee or vice-versa.
4. A maneuver to adjust the height 180° around from the maneuver point at a longitude or time.
5. A maneuver to shift the ascending node at an optimum time, longitude, time or height.
6. A maneuver to obtain a specified apogee and perigee at an optimum time, longitude, or height.
7. A maneuver to shift the ascending node and adjust the height 180° around from the maneuver point at a longitude or time.
8. A maneuver to shift the line of apsides to the maneuver point and obtain a specified height 180° around from the maneuver point at a time, longitude, or height.

The output from the GPM Processor is a display containing such maneuver information as DV, pitch, yaw, maneuver time, maneuver height, etc., and such post maneuver information as apogee and perigee heights, longitude of the ascending nodes, eccentricity, etc. A table containing the elements before and after the maneuver at the impulsive time is also output so the maneuver may be transferred to the Mission Plan Table, if desired.

3.2.2 Buttons

SET: Make an input for the GPM processor.

⏮: Move the marker down.

⏭: Move the marker up.

CLC: Calculate the maneuver.

MPT: Create finite maneuver from impulsive burn.

BCK: Go back to the main menu.

3.3 TLI Planning

3.3.1 Introduction

On a typical lunar mission all the targeting needed for TLI has been loaded into the Saturn computer (LVDC) before launch. For a dispersed Earth orbit or an alternative mission the RTCC has the capability to target a new TLI or an alternative second S-IVB burn. The options available in the TLI processor are

1. S-IVB hypersurface solution. Evaluates the preloaded targeting in the LVDC
2. Integrated free-return trajectory to the nominal lunar flyby conditions
3. Specified total Delta V
4. Specified apogee altitude
5. Integrated non-free return trajectory to the nominal nodal position at the Moon

3.3.2 Inputs

IU: Select the vessel that is used for the uplink of the updated TLI targeting. In non-MPT mode also used as the vessel from which the state vector and weights are taken for the TLI processor calculation.

MPT: Select CSM or LM mission plan table (MPT mode only).

VEC: Select the state vector used in the calculation. Normally the anchor vector of the CSM or LM ephemeris is used, but any vector in the useable vector table can be specified (MPT mode only).

Opp: Select the TLI opportunity. For option 1 this is the input to choose when the TLI is done and for all options this changes certain parameters associated with one of the opportunities in the LVDC, mostly the time of S-IVB mixture ratio shift from ignition.

Mode: Choose the TLI processor calculation mode, see list above.

TIG: Time of ignition. Not used for option 1. Used as a very rough initial guess in option 2 and 5, that only has to be accurate to a full orbit. In options 3 and 4 this is the exact time of ignition used.

DV: Delta V of the maneuver (option 3 only).

APO: Apogee altitude to be achieved by the maneuver (option 4 only).

Window: Pacific or Atlantic TLI window, for the lunar mission options (2 and 5) this is used to generate the initial guess.

SFP: For the lunar mission options (2 and 5) the target parameters are taken from the Skeleton Flight Plan table, which has two sets of data. See the Midcourse Correction Processor for more details about the SFP.

CLC: Calculate solution.

MPT: Transfer maneuver to MPT.

UPL: Uplink solution to IU.

3.3.3 Display

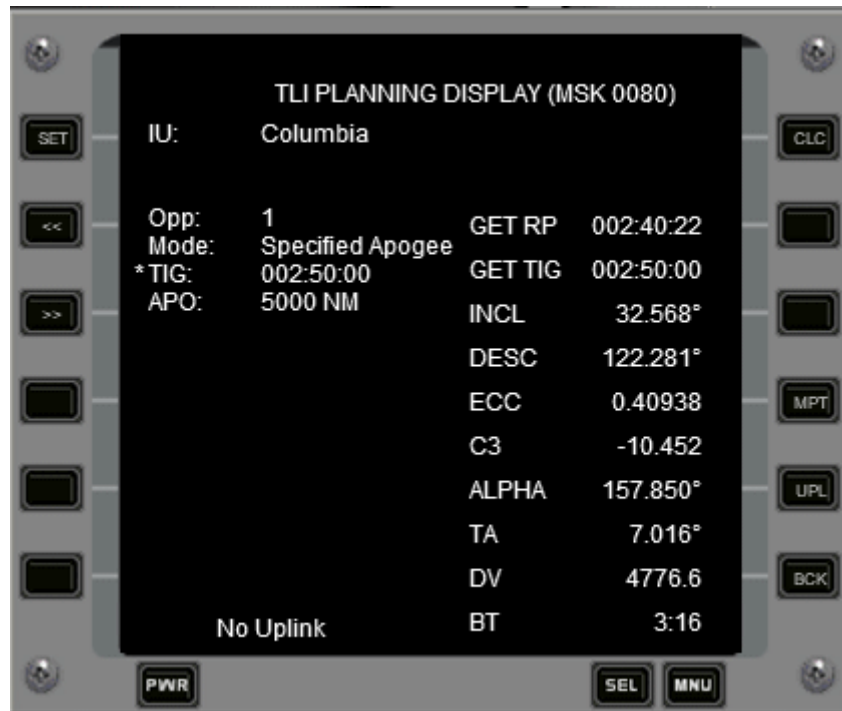


Figure 4: TLI Planning Display

Display Parameters

GET RP: Ground Elapsed Time of Restart Preparations (TB6)

GET TIG: Ground Elapsed Time of main engine ignition

INCL: Inclination at cutoff

DESC: Longitude of the descending node at cutoff

ECC: Eccentricity at cutoff

C3: Orbital energy at cutoff

ALPHA: Angle from perigee of cutoff ellipse to descending node of cutoff ellipse

TA: True anomaly at cutoff

BT: Burn time of maneuver

3.4 Midcourse Correction Processor

3.4.1 Introduction

During the translunar coast phase of an Apollo mission, it is necessary to have the capability to either correct dispersions in the nominal trajectory or determine an alternate flight plan which is within the capability of the spacecraft. This capability is provided by the midcourse correction processor. The processor has the ability to correct a dispersed state vector to some nominal end conditions, reoptimize the lunar landing mission, and generate a circumlunar flyby alternate mission. The computation types to obtain these requirements are:

1. The x, y, z, and t target update (XYZ midcourse mode).
2. The best adaptive path (BAP) reoptimization.
3. The free-return lunar flyby mode.

One or more mission options are available under each mode. The mission options, listed below, are defined by their mode, type of return, lunar parking orbit (LPO) orientation, and whether the mission is tied to a landing site.

1. X, Y, Z and T target update.
2. Free-return, fixed LPO orientation, landing site.
3. Free-return, free LPO orientation, landing site.
4. Nonfree-return, fixed LPO orientation, landing site
5. Nonfree-return, free LPO orientation, landing site
6. Circumlunar free-return flyby to nominal H_{pc} and ϕ_{pc}
7. Circumlunar free-return flyby, specified H_{pc} and nominal ϕ_{pc}
8. SPS lunar flyby to specified H_{pc} and $INCL_{fr}$
9. Optimized RCS flyby to desired or optimal inclination of free return

The MCC processor implemented in the RTCC MFD presents the last state of the processor, as used for Apollo 14 through the end of the program. Certain procedural differences for using the processor with the earlier missions arise from this, but all lunar Apollo missions are fully supported.

3.4.2 Input/Output

Inputs for all midcourse modes fall into two categories: those from the data table (also called skeleton flight plan), and those which are manually entered during the mission by the user. The data table contains variables which are needed to execute the different options. These variables may be target parameters used in the XYZ and T mode or first guesses for certain variables. The table also contains parameters which change according to the nominal mission design and launch day (e.g., the lunar landing site). Output parameters from a BAP midcourse can be used to update the data table for later midcourse calculations or the XYZ and T midcourse mode. The data table and the manual inputs

are defined in table I and II.

Output from the MCC program are of three types: those displayed, those that are needed for executing the midcourse maneuver, and those which update the data table. Displayed parameters are shown in table III. Output for the data table is shown in table I. BAP's are the only options that update the data table.

In the RTCC MFD the MCC processor consists of three display pages: computational inputs, constraints on the solutions and the MCC tradeoff display (modelled after the real display used in Mission Control Houston). On the constraints page some of the inputs are made in the MED format (Manual Entry Device), which is the same format as was used in the real RTCC. The MEDs for the midcourse processor all have codes starting with an F, e.g. F22. The MED inputs are checked for errors and certain omissions are replaced by default values.

3.4.3 SFP page

The SFP page contains the skeleton flight plan table (SFP) parameters that are inputs and outputs for the midcourse processor. Usually only one operation has to be done on this page for a nominal mission. Before any midcourse maneuver can be calculated, the initial SFP (Called the preflight table) has to be interpolated from tables that are valid for the whole daily launch window. This is accomplished by using the F62 button and entering the TLI opportunity (1 or 2) and the launch azimuth (usually but not always 72 degrees). The input format is: F62,,1,72.0; The launch day is an optional parameter that could be entered after the first comma, but is usually not needed. Once the F62 MED was entered the SFP table should populate with numbers. Afterwards the MCC processor can be used for calculations.

3.4.4 Computation page

MAN: Maneuver option (1 to 9), see description above.

VTI: Vector time. Time tag of the state vector from the ephemeris (mission planning mode only).

IG: Impulsive time of ignition of the midcourse maneuver.

COL: Column for the solution. The Midcourse Tradeoff display can hold up to 4 different solutions from the MCC processor.

CFG: Configuration for the midcourse maneuver, options are docked or undocked. Used to calculate certain display parameters only.

SFP: Skeleton flight plan (data table) used for initial guesses and target parameters. Table 1 usually contains the preflight data, table 2 the results of a previous BAP midcourse calculation that were transferred from the midcourse tradeoff. Tables 3 to 5 will be supported in the future.

MID: Go to midcourse tradeoff display MFD page.

HPC: Pericyynthion altitude for lunar flyby modes.

INC: Free return inclination for lunar flyby modes 8 and 9. Mode 9 is further divided into mode 9A (RCS optimized flyby) and 9B (SPS optimized flyby to specified free return inclination). If the input inclination is 0, then mode 9A, the fully optimized flyby, will be used. Otherwise the specified inclination is attained. By using a plus or minus sign for the inclination an ascending or descending return can be specified (travelling from south

to north and north to south at reentry respectively).

CON: Go to midcourse constraints MFD page.

BCK: Go back to previous menu.

3.4.5 Constraints page

F22: Azimuth constraints. Input method: "F22, Minimum Azimuth,Maximum Azimuth;" Limited to -110° to -70° . Used by modes 3 and 5. Constrains the approach azimuth to the landing site at the time of landing. Special logic is used if the min and max azimuths are set the identical. In that case the lunar orbit has a fixed orientation, although without imposing the LOI/DOI geometry. This should be done for missions which used the LOI-1/LOI-2 maneuver sequence (Apollo 8,10-12). Example: F22,-90,-90;

F23: Time constraints. Input method: "F23,TLMIN,TLMAX;" Used by modes 4-5. This sets a minimum or maximum time limit for the arrival at pericynthion. Useful for missions with stricter timing requirements for arriving in lunar orbit (Apollo 14 to 17). If omitted (input: "F23;") The constraints are zeroed and the pericynthion time is not constraint.

F24: Reentry constraints. Input method: "F24,Flight Path Angle,Reentry Range;" Used in the free return and lunar flyby modes. Inputs are the flight path angle at entry interface and the range from entry interface to landing.

F29: Pericynthion height limits. Input method: "F29,HPMIN,HPMAX;" Used in mode 9 only. Can be used to force the solution indirectly to a different splashdown longitude.

LAT: Latitude bias for modes 8 and 9. TBD

INC: Maximum inclination for the powered return (TEI). Not enforce yet.

LOI: Apolune and perilune height of the LOI (LOI-1) ellipse.

DOI: Apolune and perilune height of the DOI (LOI-2) ellipse.

REV: Input: REVS1 REVS2. Number of orbits spent in the first (LOI to DOI/LOI-2) and second (DOI/LOI-2 to landing site) lunar orbit. REVS2 is always an integer, REVS1 can contain partial orbits.

ROT: Input: SITEROT ETA. The first parameter is the true anomaly at the landing site at the time of landing. Usually PDI should happen at perilune, which will be 15° ahead of the landing site. In that case 15 should be the input. ETA is the true anomaly of LOI on the post LOI orbit. This will usually be consistent with the REVS1 parameter, which will put DOI at perilune.

PC: Revolutions before and after the lunar orbit plane change maneuver. Used to estimate the trajectory in lunar orbit. The first parameter M is the number of orbits between the lunar landing and the plane change maneuver. The parameter N is the number of orbits between the plane change and lunar ascent.

BCK: Back to midcourse calculation page.

3.4.6 Midcourse Tradeoff Display

COLUMN	1	2	3	4
MODE	3	5	9	
RETURN	FREE	NONFREE	FREE	
AZ MIN	-91.000	-91.000	0.000	
AZ MAX	-91.000	-91.000	0.000	
WEIGHT	98890	98890	98890	
GETMCC	026:59:59	026:59:59	026:59:59	
DV MCC	19.9	10.7	11.6	
YAW MCC	0.000	0.000	0.000	
H PYCN	61.571	59.478	80.812	
GET LOI	075:58:44	076:11:10	000:00:00	
DV LOI	2908.5	2910.7	0.0	
AZ ACT	-91.004	-91.004	0.000	
I FR	34.801	0.000	29.618	
I PR	52.116	52.188	0.000	
V EI	36211.6	36203.8	36153.3	
G EI	-6.520	-6.520	-6.520	
GETTEI	136:33:20	135:47:23	000:00:00	
DV TEI	3330.8	3297.8	0.0	
DV REM	0.0	0.0	0.0	
GET LC	195:05:30	195:04:30	145:44:27	
LAT IP	11.309	11.315	-2.283	
LNG IP	-170.106	-170.095	-175.056	
DV PC	16.5	16.5	0.0	

Figure 5: Midcourse Tradeoff Display

COLUMN: Shows up to 4 midcourse correction solutions.

MODE: Shows the mode (1-9) that was calculated.

AZ MIN: Minimum approach azimuth at the landing site.

AZ MAX: Maximum approach azimuth at the landing site.

WEIGHT: Weight at ignition in lbs.

GETMCC: Estimated time of ignition of the midcourse correction (actual, not impulsive).

DV MCC: Total DV of the midcourse correction in feet per second.

YAW MCC: Yaw angle (out-of-plane) of the maneuver.

H PYCN: Height of pericyynthion resulting from the maneuver.

GET LOI: Estimated time of ignition of LOI (actual, not impulsive).

DV LOI: Total DV of LOI maneuver.

AZ ACT: Actual approach azimuth at the landing site.

I FR: Free return inclination, Earth referenced.

I PR: Powered return (TEI) inclination, Earth referenced.

V EI: Velocity at entry interface in feet per second.

G EI: Flight path angle (gamma) at entry interface.

GETTEI: GET of the TEI maneuver.

DV TEI: Total DV of the TEI maneuver.

DV REM: DV remaining after TEI (not implemented).

GET LC: GET of splashdown.

LAT IP: Latitude of splashdown (impact point).

LNG IP: Longitude of splashdown (impact point).

DV PC: Total DV of lunar orbit plane change maneuver.

3.5 Lunar Orbit Insertion (LOI) Processor

3.5.1 Introduction

The LOI processor calculates the LOI-1 maneuver for an Apollo lunar mission. The maneuver is targeted based on the following assumed trajectory profile to the landing site. All plane change is accomplished with the first burn. A second burn (LOI-2 or DOI) adjusts the inplane orbital elements so that a specified orbit occurs at the landing site. It is not not always possible to meet all desired end conditions; thus various solutions are computed.

There are four solution types, each with a positive-negative solution, for a total of eight solutions. A positive solution is one whose perilune is rotated ahead (i.e., in the direction of motion); a negative solution is one whose perilune is rotated behind (i.e., opposite to the direction of motion). The four types are as follows.

- Plane solutions: obtain the desired azimuth at the landing site, giving up the lunar orbit perilune if necessary, which is if the node between the incoming trajectory (approach hyperbola) and the orbit after LOI occurs at an altitude below the desired perilune, or above the desired apolune. This is the type of LOI maneuver generally used for Apollo 12 and earlier.
- Coplanar solutions: obtain the desired lunar orbit shape (apolune and perilune) in the plane of the approach hyperbola with a pre-hyperbolic perilune impulsive point for the positive solution and a post-hyperbolic perilune impulsive point for the negative solution. This solution type therefore has no plane change.
- Minimum Theta solutions: obtain the desired lunar orbit shape (apolune and perilune) and minimize the wedge angle between the actual and desired lunar orbit plane within an input maximum allowable DV.
- Intersection solutions: adjusts the first lunar orbit perilune altitude to obtain a specified altitude difference (or intersection with no altitude difference) between it and the altitude on the post-DOI lunar orbit. This is the type of LOI maneuver generally used for Apollo 13 and later and has no use if the second lunar orbit maneuver (LOI-2) is a circularization maneuver.

The LOI processor implemented in the RTCC MFD is based on the one used for Apollo 14 and later. Most capabilities of earlier programs are retained, so that all lunar missions are still supported.

3.5.2 Inputs

The inputs for the LOI processor are divided in initialization parameters and computation parameters.

Computation Parameters

INI: Got to LOI initialization page.

VTI: Time for taking the state vector from the CSM ephemeris (MPT mode only).

APO: Apolune height after LOI.

PER: Perilune height after LOI.

DVP: Maximum DV for positive Min Theta solution.

DVN: Maximum DV for negative Min Theta solution.

DIS: Got to LOI display.

AMN: Choose the minimum approach azimuth to the landing site.

ADS: Choose the desired approach azimuth to the landing site.

AMX: Choose the maximum approach azimuth to the landing site.

BCK: Back to main menu.

Initialization Parameters

HA: Apolune height after DOI/LOI-2.

HP: Perilune height after DOI/LOI-2.

DW: Angle of perilune from the landing site (negative if the landing site is post-perilune).

R1: Number of revolutions in the first lunar orbit (may have a fractional part).

R2: Number of revolutions in the second lunar orbit.

ETA: True anomaly of LPO-1 for transferring from the hyperbola to LPO-1.

DHB: Altitude constraint of the intersection solutions. The bias is negative if LPO-2 is to be below the LPO-1 perilune.

PLA: A flag to specify if plane or minimum Theta nodes should be used to compute intersection solutions.

BCK: Back to LOI computation page

3.5.3 LOI Display

This display is based on the actual display used by the flight controllers for Apollo 14.

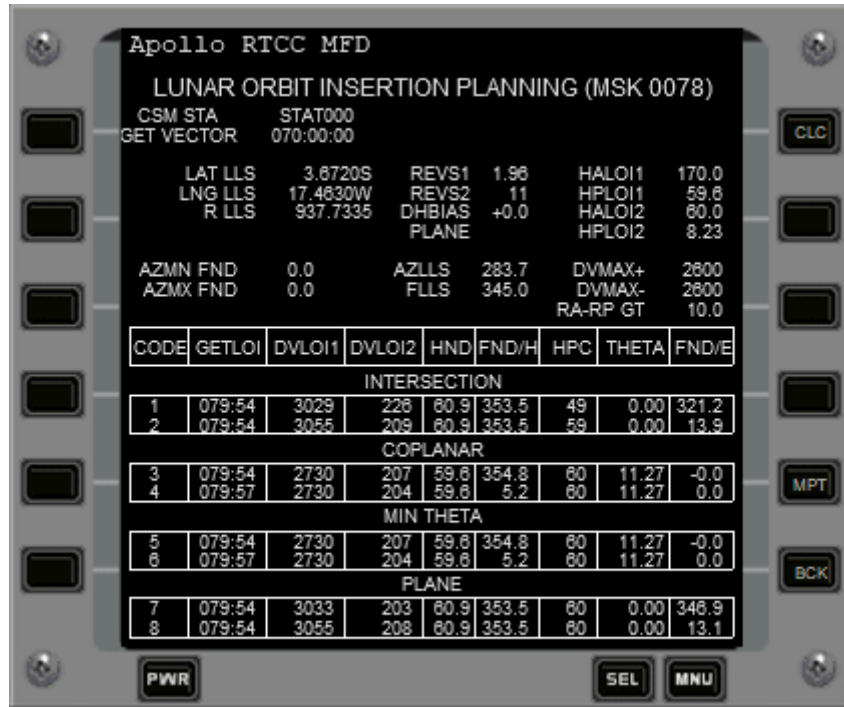


Figure 6: LOI Planning Display

Display Parameters

CSM STA: Station ID of the state vector used to target the maneuvers (not implemented).

GET VECTOR: GET of state vector used to target LOI.

LAT LLS: Latitude of the landing site in degrees.

LNG LSS: Longitude of the landing site in degrees.

R LLS: Radius of the landing site in nautical miles.

REVS 1: Revolutions in LPO-1.

REVS 2: Revolutions in LPO-2.

DH BIAS: Height bias for the intersection solutions.

AZ LLS: Desired azimuth at the landing site.

FLLS: Angle of perilune from the landing site.

HALOI1: Apolune height on first lunar orbit.

HPLOI1: Perilune height on first lunar orbit.

HALOI2: Apolune height on second lunar orbit.

HPLOI2: Perilune height on second lunar orbit.

DVMAX+: Maximum allowable DV for the positive Min Theta solution.

DVMAX-: Maximum allowable DV for the positive Min Theta solution.

RA-RP GT: Tolerance for the calculation of DVLOI2 in nautical miles.

CODE: Code for the eight possible solutions.

GETLOI: Impulsive GET of LOI ignition.

DVLOI1: Total DV of LOI-1 in feet per second.

DVLOI2: Total DV of DOI/LOI-2 in feet per second.
HND: Height of the node (impulsive LOI ignition point).
FND/H: True anomaly at LOI on the approach hyperbola (pre LOI).
HPC: Height of perilune on the first lunar orbit.
THETA: Angle between the desired lunar orbit plane and the actual achieved plane.
FND/E: True anomaly at LOI on the first ellipse (post LOI).

3.6 Lunar Descent Planning Processor

The LDPP computes maneuver sequences to be executed by the CSM and the LM to yield a set of desired orbital conditions at the point of LM descent ignition or a desired CSM orbital plane at the time of LM liftoff. There are seven modes of operations from which to choose. Each of these modes may contain more than one sequence. Each sequence may or maybe not have more than one maneuver. The following table presents a list of maneuver sequences in each mode:

Mode	Maneuver	Sequence				
		1	2	3	4	5
1	1	PC	PCC	ASP	PCCH	PCCT
	2	DOI	DOI	CIA	DOI	DOI
	3	—	—	DOI	—	—
2	1	ASH	CIR	—	—	—
	2	DOI	DOI	—	—	—
3	1	ASH	—	—	—	—
	2	CIA	—	—	—	—
	3	DOI	—	—	—	—
4	1	DOI	—	—	—	—
5	1	PC	HO1	HO1	—	—
	2	HO1	PC	HO2	—	—
	3	HO1	HO2	PC	—	—
	4	DOI	DOI	DOI	—	—
6	Go to powered descent				—	—
7	1	PPC			—	—

Abbreviations:

ASP	Apside shift, altitude adjustment, and plane change maneuver
ASH	Apside shift and altitude adjustment maneuver
CIA	Circularization at an apsis maneuver
CIR	Circularization maneuver
DOI	Descent orbit injection maneuver
HO1	First maneuver in a double Hohmann sequence
HOI2	Second maneuver in a double Hohmann sequence
PC	Plane-change maneuver
PCC	Plane change and circularization maneuver
PCCH	Plane change and circularization maneuver at a desired height
PCCT	Plane change and circularization maneuver at a desired time
PPC	Prelaunch plane-change maneuver

Detailed description of each mode:

Maneuver Mode No.	Maneuver Sequence No.	Maneuver Display Code	Definition
1	1	PC	Compute plane-change maneuver only
1	2	PCC	Compute plane-change and circularization maneuver
1	3	ASP-CIA	Compute plane-change maneuver combined with first maneuver of a CSM two maneuver sequence to circularize the CSM orbit at an input altitude
1	4	PCCH	Compute plane-change and circularization maneuver at a specific altitude
1	5	PPCT	Compute plane-change and circularization maneuver at an input time
2	2	ASH	Compute CSM maneuver to establish an apsis and an input altitude at the DOI maneuver point
2	3	CIR	Compute CSM maneuver to circularize orbit at an input altitude
3	1	ASHT-CIA	Compute CSM two-maneuver sequence with the first maneuver performed at an input time and the second maneuver performed at an input altitude to circularize the orbit
3	3	ASHA-CIA	Compute CSM two-maneuver sequence with the first maneuver performed at an apsis and the second maneuver performed at an input altitude to circularize the orbit

Maneuver Mode No.	Maneuver Sequence No.	Maneuver Display Code	Definition
4	1	DOI	Computes the DOI maneuver only
4	2	DOI	Computes the DOI maneuver only and includes a plane change
4	3	DOI	Computes the DOI maneuver with a higher fidelity numerical integration processor, for a more precise perilune altitude and placement. Useful for CSM DOIs that take place many hours before PDI.
4	4	DOI	Same as sequence 3, but with an included plane change.
5	1	PC, HO1, HO2	Compute CSM three-maneuver sequence so that the first maneuver is a plane change and the following pair is a double Hohmann to a circular orbit at an input altitude
5	2	HO1, PC, HO2	Compute CSM three-maneuver sequence so that the first maneuver initiates a double Hohmann, the second is a plane-change, and the third completes the double Hohmann to a circular orbit at an input altitude
5	3	HO2, HO2, PC	Compute CSM three-maneuver sequence so that the first two maneuvers constitute a double Hohmann to a circular orbit at an input altitude and the third is a plane change
6	1	PDI	Powered descent only
7	1	PPC	Prelaunch plane change

3.6.1 Initialization Parameters

Button	Parameter	Description
AZI	Approach azimuth	For the calculation modes that involve a plane change maneuver, the approach azimuth to the landing site can be specified. If this is not required then the parameter can be set to 0, which will result in the optimum (smallest) plane change burn to be calculated
HDP	Altitude of point of descent ignition	Usually the DOI maneuver will set up PDI to occur at this specific altitude
PDS	Powered-descent simulation flag	Set to perform a a detailed simulation of powered descent (not implemented yet)
PDI	Powered-descent time	If the powered descent is to be simulated, this option either lets you choose the time of PDI or let it be internally calculated. For this the time should be set to zero
N	Dwell orbits	Number dwell orbits desired between DOI and powered-descent ignition
DFT	Descent flight time	Time from PDI to touchdown
DFA	Descent flight arc	Travel angle from PDI to touchdown. Usual values are 15 to 16 degrees
LSO	Landing site offset	Usually this will be set to the same value as DFA. This will cause PDI to occur at perilune. If PDI is desired to occur at a different point in the orbit, like on Apollo 17, then this value can control where the perilune will be located relative to the landing site. A value of -10 degrees will have the perilunar location 10 degrees after landing site passage

3.6.2 Computation Parameters

VEH	Vehicle	Select if CSM or LM are performing the maneuver sequence
VTI	Vector time	With MPT enable, this time is the vector time from the ephemeris. Otherwise it seconds as a vehicle selection button
MOD	Mode	Select the calculation mode (1-7)
SEQ	Sequence	Select the specific maneuver sequence
HEI	Height	Altitude wanted at apsis. Typically the altitude at which DOI is desired to occur, but depending on the mode can also be the circularization maneuver altitude
TH1	Threshold time 1	Threshold time for the first maneuver of a sequence
TH2	Threshold time 2	Threshold time for the second maneuver of a sequence
TH3	Threshold time 3	Threshold time for the third maneuver of a sequence
TH4	Threshold time 4	Threshold time for the fourth maneuver of a sequence. DOI maneuvers will always use this time as the threshold for the maneuver. For a prelaunch plane-change maneuver this time is used as a threshold for overflying the landing site

3.6.3 Display

Buttons:

CLC: Calculate the display

TLA: Store the time of landing (GETTD) for RTCC wide use

MPT: Convert to finite burn or transfer to MPT

Parameter	Description	Unit
LM WT	LM weight	lbs
GMTV	GMT of state vector used in the calculation	
GETV	GET of state vector used in the calculation	
MODE	Calculation mode of the calculation	
LAT LLS	Latitude of the landing site	degrees
LONG LLS	Longitude of the landing site	degrees
MVR	Maneuver code	
GETTH	Threshold time for the maneuver	
GETIG	Time of ignition for the maneuver. Special case for the PPC mode: Second time is the landing site crossing	
LIG	Longitude of ignition	degrees
DV	Total Delta V of the maneuver	ft/s
HAC	Height of apolune after the maneuver	NM
HPC	Height of perilune after the maneuver	NM
DVX	Delta V x-component of the maneuver in LVLH coordinates	ft/s
DVY	Delta V y-component of the maneuver in LVLH coordinates	ft/s
DVZ	Delta V z-component of the maneuver in LVLH coordinates	ft/s
GETIG	Time of ignition of powered descent	
GETTD	Touchdown time	
MODE	Descent azimuth mode	
DESC AZ	Approach azimuth to the landing site, if input	degrees
SN.LK.A	Sun look angle at the time of landing	degrees

Error codes:

Bit	Description
1	Unrecoverable AEG error. Processing halted.
2	Error in backing through maneuver(s) in PCPLCH. Processing halted.
3	Error in trying to obtain LM weight from PLAWDT (or LM weight is zero). Processing continued.
4	Failed to converge in ASH maneuver. Processing continued.
5	Failed to converge on closest approach in PCPLCH. Processing continued.
6	Failed to converge on landing site. Processing continued.
7	Failed to converge in DOI maneuver. Processing continued.
8	Failed to converge on a height maneuver in PCSAC. Processing continued.
9	Failed to converge on a common node. Processing continued.
10	Delta V for plane change maneuver less than 1 ft/s. No maneuver computed, processing continued.
11	Error from PMMAPD. Processing halted.
12	Failed to converge on time of landing site longitude crossing PCTLLC - Processing continued.

3.7 Lunar Launch Window Processor

3.7.1 Introduction

The Lunar Launch Window Processor (LLWP) computes the lunar module recommended time of lift-off from the lunar surface and the lunar launch window. It is used for the coelliptic rendezvous profile with the CSI and CDH maneuvers. For the short profile (Apollo 14 and later), see the lunar launch targeting processor.

The LLWP computes lift-off times as a function of the height difference (Δh) in the coelliptic orbits phase of the lunar concentric rendezvous plan. The recommended LM time of lift-off from the lunar surface is that launch time during a given CSM revolution for which the required Δh is the nominal mission value. The lunar launch window is defined as the total interval of time in the CSM revolution during which the IM can lift-off and rendezvous with the CSM within the limits of maneuver- ΔV budgets, LM ascent stage power lifetime, and safe orbital altitudes. The lift-off time computation involves simulation of the mission from LM launch through rendezvous. The lift-off time which satisfies phase convergence for the given coelliptic Δh at terminal phase initiation is the required time. The LLWP output for each Δh includes the lift-off time, maneuver times, and maneuver- ΔV values.

The LLWP times may be computed based on either LM or CSM execution of the rendezvous maneuvers. In either case, Δh is positive when the maneuvering vehicle is below the target vehicle, and is negative otherwise. The positive Δh values are limited by the minimum allowable pericynthion altitude. The negative Δh values are limited by the rendezvous ΔV budgeted for the maneuvering vehicle; however, a theoretical limit exists above which the vehicle cannot acquire the target along the input elevation angle.

The MFD has two initialization pages, corresponding to MEDs K13 and K14. Because LGC P12 targets and powered flight arc and time are inputs, it can improve the accuracy of the calculation to simulate through a lunar ascent with an updated liftoff time. After a LLWP calculation go to the lunar ascent page, input the desired liftoff time and run the ascent simulation. This updates the actually required powered flight arc and time which can then be used in another LLWP calculation.

3.7.2 Initialization

The following inputs correspond to MED K13, Initialization for LM Targeting and Launch Window:

PFA: Powered flight arc from lunar liftoff to insertion, in degrees.

PFT: Powered flight time from lunar liftoff to insertion, in seconds.

HEI: Height of insertion, in feet.

VLH: Horizontal velocity at insertion, in feet per second. Corresponds to LGC P12 input.

VLV: Vertical velocity at insertion, in feet per second. Corresponds to LGC P12 input.

YAW: Maximum yaw steering capability during ascent, in degrees. Effectively a maximum crossrange.

LIF: Maximum ascent stage lifetime since launch, in hours.

MHE: Minimum safe height from rendezvous maneuvers, in nautical miles.

LDV: Maximum LM Delta V capability, in feet per second.

CDV: Maximum CSM Delta V capability, in feet per second.

The following inputs correspond to MED K14, Launch Window Initialization:

BIA: Time bias for CSI if CSM is active vehicle. CSI will be done at LM apsis plus input time bias. In minutes.

E: Elevation angle to initiate terminal phase (TPI) on, in degrees.

WT: Terminal phase travel angle from TPI to TPF, in degrees.

TPH: Offset height difference at TPF, in nautical miles.

TPP: Phase angle offset at TPF, in nautical miles.

DHB: Height difference to begin curve fit, in nautical miles.

DHI: Height difference increment for curve fit, in nautical miles.

3.7.3 Calculation

These inputs correspond to MED K15, Generate Launch Window:

CHA: Choose either CSM or LM as the chaser, the active vehicle in the rendezvous.

VTI: Choose the CSM vector time for the LLWP calculation. MPT mode only.

THT: Either a threshold time for liftoff or the TPI time if it was input. Currently only the second option is implemented.

CSI: CSI maneuver flag. If 0 is input then CSI is done 90 degrees from insertion. If any negative value is input then CSI is done at apolune after insertion. If a positive value is input then CSI is done at a delta time from insertion. Typically CSI is set at 50 minutes after insertion.

CDH: CDH maneuver flag. If 0 is input then CDH is done at an upcoming apsis after CSI. If positive, it must be an odd number and CDH is done at N/2 orbits after CSI.

TGT: Select the target vehicle, typically the CSM (non-MPT mode only).

TPI: TPI definition. Enter the desired TPI time. This must be generated externally, either with the TPI times page or the sunrise/sunset table (MPT mode only).

LLW: Delta height flag. A positive number indicates request for complete launch window and the number of delta heights. A negative number indicates request for the input delta heights only. Presently only the second option is implemented.

DH: Input three delta heights, being used if the delta height flag indicates delta heights are being input.

3.7.4 Display



The image shows a screenshot of the Apollo RTCC MFD (Mission Flight Display) interface. The display is titled "Apollo RTCC MFD" and "LUNAR RENDEZVOUS PLAN TABLE". It contains several tables of data related to the lunar rendezvous mission. The top section shows CSM and LM parameters, including position, time, and life support. The bottom section shows a table of delta heights (DH) for three different scenarios (ID 1, 2, and 3).

CSM STA		LM POSITION		LM STA		LM LIFE	
GMTV	φLLS +0.7147	GMTV		DV MAX	9.0 h		
GETV	ΔLLS +23.7085	GETV		MIN H	430		
MVR VEH	LEM	THT	127:05:30	DTCSI	000:50:00	WT	
ID	M	DH	GETLO GETINS	GETCSI DVCSI	GETCDH DVCDH	GETTPI DVTPI	DVTPF DVT
1	2	10.0	124:30:29 124:37:44	125:27:44 56.8	126:26:02 6.3	127:05:30 16.3	21.5 100.9
2	2	15.0	124:31:10 124:38:25	125:28:25 50.0	126:26:31 4.8	127:05:30 24.6	31.6 111.0
3	2	20.0	124:31:52 124:39:07	125:29:07 43.1	126:26:59 10.0	127:05:30 33.1	41.3 127.5

Figure 7: Example LLWP Calculation

Display Parameters

CSM STA: Station ID of CSM state vector used in the LLWP calculation.

GMTV: GMT of CSM state vector.

GETV: GET of CSM state vector.

MVR VEH: Active vehicle, performing the rendezvous maneuvers.

LM POSITION: Latitude and longitude of the lunar landing site, from which the LLWP simulates the launch.

THT: Threshold time or TPI time.

LM STA: Station ID of LM state vector used in the LLWP calculation.

GMTV: GMT of LM state vector.

GETV: GET of LM state vector.

DTCSI: Input time from insertion to CSI.

LM LIFE: LM maximum life time in hours.

DV MAX: Maximum DV of active vehicle.

MIN H: Minimum safe height for rendezvous maneuvers, in nautical miles.

ID: Number in table of solutions.

M: Number of revolutions from launch to TPI.

DH: Delta height of solution, in nautical miles.

GETLO: GET of calculated liftoff time.

GETINS: GET of calculation insertion time.

GETCSI: GET of calculated CSI time.

DVCSI: Delta V of CSI maneuver.

GETCDH: GET of calculated CDH time.

DVCDH: Delta V of CDH maneuver.

GETTPI: GET of calculated TPI time.

DVTPI: Delta V of TPI maneuver.

DVTPF: Delta V of TPF maneuver.

DVT: Total Delta V of rendezvous plan.

3.8 Deorbit Targeting

3.8.1 Introduction

The retrofire targeting calculates the time of ignition and the burn parameters for a maneuver to return the spacecraft back to Earth from Low Earth Orbit. This can include a separation maneuver from the S-IVB at a fixed time before the deorbit burn. Or a shaping maneuver, that shapes the orbit before reentry, and happens at a given GET.

In the MFD there are pages for the definition of the separation/shaping maneuver and the retrofire maneuver respectively. To select a splashdown target and estimated time of ignition the Recovery Target Selection Display has been implemented. Finally a page to specify the type of target of the deorbit burn can be used.

3.8.2 Separation/Shaping Inputs

SHA: Time of ignition for the shaping maneuver. If no shaping maneuver is desired this should be set to zero.

SEP: Time in minutes that the separation maneuver will occur before the deorbit burn.

THR: Engine for the sep/shaping maneuver. Options are the SPS or the SM RCS.

DV: Fixed Delta V of the sep/shaping maneuver. Enter either a DV or a burn time, but not both.

DT: Fixed burn time of the sep/shaping maneuver. Enter either a DV or a burn time, but not both.

ATT: Attitude in LVLH coordinates of the sep/shaping maneuver. The default values are for retrograde, heads up, 31.7° window line on the horizon.

UDT: Duration of the ullage burn for the sep/shaping maneuver. Only applies to SPS maneuvers.

UTH: Number of ullage thrusters (2 or 4). Only applies to SPS maneuvers.

GBL: Calculate the gimbal angles or use system parameters. Only applies to SPS maneuvers.

3.8.3 Retrofire Constraints

ENG: Engine for the retrofire maneuver. Options are the SPS or the SM RCS.

MOD: Choose the burn mode for the retrofire maneuver. The options are a fixed Delta V, a fixed burn time or a maneuver to reach the reentry target line (velocity vs. flight path angle at entry interface).

VAL: Desired Delta V or burn time for the retrofire maneuver. Doesn't apply to velocity-flight path angle targeting.

ATT: Attitude mode for the retrofire maneuver. Either an input LVLH attitude or the 31.7° window line on the horizon.

LVH: LVLH attitude for the retrofire maneuver.

ULL: Ullage options for the retrofire maneuver. Only applies to the SPS.

GIM: Calculate the gimbal angles or use system parameters. Only applies to SPS maneuvers.

K1: Initial bank angle for the reentry. Held from entry interface to the specified G-Level.

GC: G-Level at which the final bank angle is started to be used.

K2: Final bank angle for the reentry. Held from the specified G-Level to drogue chute opening. In the case of target a specified latitude as well as longitude the bank angle will be reversed.

BCK: Back to deorbit targeting menu.

3.8.4 Target Selection

This program computes and displays groundtrack data for any requested 40 degrees of longitude for a requested time and starting longitude.

CLC: Calculate recovery target selection display. Inputs are a threshold time and the starting longitude.

PAG: Cycle through the pages of the display.

SEL: Choose one of the sets of coordinates on the groundtrack and save the latitude, longitude and an estimated GET for the retrofire maneuver to be used for the retrofire computation.

BCK: Back to deorbit targeting menu.

3.8.5 Retrofire Calculation

TYP: Choose the type of calculation. The options are type 1 (no sep/shaping maneuver) or type 2 (with sep/shaping maneuver).

GET: Estimated time of ignition for the retrofire maneuver.

LAT: Desired splashdown target latitude. Set this to a large negative value to disable targeting the latitude. The reentry profile is then a fixed bank angle and not a bank, reverse bank angle.

LNG: Desired splashdown target longitude.

MD: Maximum miss distance of the splashdown target. Can be set to a larger value to improve convergence.

DIG: Go to Retrofire Digitals display.

XDV: Go to Retrofire External DV display.

SEP: Go to Retrofire Separation display.

BCK: Back to deorbit targeting menu.

3.9 Return-to-Earth Targeting

3.9.1 Introduction

The Return-to-Earth targeting calculates maneuvers for returning the spacecraft back to Earth from beyond Low Earth Orbit. For LEO maneuvers see the deorbit targeting. The objective is to calculate a single maneuver for changing the trajectory to one having safe entry-interface conditions and satisfying certain other constraints. Other constraints are dependent on the type of abort requested. The safe entry interface condition is a velocity flight path angle target line at 400,000 feet. Either of two entries may be specified, a shallow or a steep target line. The steep target line was used for all lunar Apollo missions.

Dependent upon the request, the abort maneuver will be computed considering one of three types of impact areas:

1. An Unspecified Area
This means that a safe reentry is guaranteed but no consideration is given to the location of the impact point (it may be quite undesirable).
2. A Primary Target Point (PTP)
This would be defined by a pair of latitude-longitude values. Not currently implemented in NASSP.
3. An Alternate Target Point (ATP)
This is actually not a point but is as many as five connected line segments (defined by latitude-longitude pairs) extending generally in a longitudinal direction.

Both the PTP and ATP targets can be defined by manual input. The Return-to-Earth Target Table Display contains up to five PTP and five ATP target names and definitions.

Three general types of abort maneuvers are available.

1. Time Critical Unspecified Area (TCUA)
This is an inplane maneuver producing the trajectory with the earliest possible reentry. The abort is characterized by
 - (a) Consuming all fuel provided for the maneuver, or
 - (b) Having the maximum allowable reentry velocity, or
 - (c) Having the minimum allowable time from maneuver to entry to Entry interface.
2. Fuel Critical Unspecified Area (FCUA)
This is an inplane maneuver requiring the least fuel to obtain a safe entry interface. Not that if the pre-abort trajectory had a safe entry interface, a request for a FCUA abort having the same target line should result in a zero DV maneuver. In practice, because of convergence tolerances, a small maneuver will be computed.
3. ATP Abort
This inplane maneuver produces a trajectory that impacts ATP trace. Multiple solutions are possible. If they exist, they are discrete and differ by twenty-four hour increments in time of landing.

The abort computations are separated into three distinct steps. They are (1) the trade-off, (2) the Abort Scan Table, and (3) return-to-Earth digitals. Associated with each step in the computations is a display which summarizes the results of that step. The construction of an abort solution may be viewed as combining models of the abort maneuver, the trajectory between maneuver and entry interface, and the reentry trajectory. The three are combined in a way that will satisfy the abort criteria and the solution constraints. The trade-off, Abort Scan Table, and return-to-Earth digital computations are characterized by the models used; the models progress from less precise to more precise solutions.

3.9.2 Tradeoff

Four major questions must be answered in abort planning:

1. How much fuel may be expended?
2. When may the maneuver be performed?
3. How soon must splashdown occur?
4. If the target is a PTP, how large a miss is acceptable?

There is a hidden difficulty in answering these questions — the best answer to any one of the above is not independent of the answers to the remaining three. Sometimes by relaxing the acceptable miss by a few miles, the required velocity decreases by several thousand feet per second; sometimes shifting the time of the maneuver by a few minutes substantially reduces the time of landing or the chance of a miss.

The objective of this step of the computations is to provide the user with an overall picture of the abort situation for either ATP or PTP aborts. The abort situation is defined in terms of the above four parameters (three if the target is an ATP). To this end the user supplies parameter ranges within which his solutions of interest can be found. The RTE section imposes a mesh over the solution region and examines each mesh point for a solution. Those solutions existing are used to produce analog TV displays to assist the user in arriving at an optimum answer to the four questions given above.

As mentioned earlier, multiple solutions having different times of landing may be available for ATP and PTP aborts. As the time of abort is varied, these times of landing tend to vary smoothly where they exist. This leads to a natural grouping of solutions into families of solutions, each having similar times of landing. The Tradeoff Display is a multiple page (up to five) TV display; each page contains the analog information for a different family of solutions. Two display formats are used.

1. The Remote Earth ATP consisting of one analog graph, characteristic DV versus Time of Abort
2. Near-Earth ATP consisting of three analog graphs on each page:
 - (a) Characteristic DV versus Time of Abort
One curve representing the DV required for an inplane solution that impacts the ATP trace.
 - (b) Time of Landing versus Time of Abort
One time of landing curve corresponding to the curve in item a.

(c) Latitude of Impact versus Time of Abort

One curve representing the declination of impact. Although latitude is not considered a tradeoff parameter, it is helpful to know this information.

The abort solutions for the Tradeoff Display are constructed by combining an impulsive velocity change, a conic (analytic two-body) trajectory, and polynomial reentry functions. The reentry functions simulate either a high-speed G&N entry or a constant G-level entry and are valid only if entry interface occurred on one of the two target lines. They provide reentry range, cross range, and DT as a function of target line, entry profile and entry interface velocity, inclination, azimuth, and latitude. Solutions constructed for the Abort Scan Table or return-to-Earth digitals will be constrained to these conic entry interface conditions.

Currently the tradeoff display only works in Earth reference.

Inputs

MOD: Switch between near-Earth or remote Earth format.

REM: Choose the page (out of 5) for the remote Earth solution to be displayed

SIT: Choose the tradeoff site from the target table.

TV: Choose the vector time (MPT mode only).

MIN: Choose the minimum abort time.

MAX: Choose the maximum abort time.

PAG: Cycle between the tradeoff pages.

CLC: Calculate tradeoff solution.

ENT: Choose the entry profile.

3.9.3 Abort Scan Table (AST)

The Abort Scan Table Display is essentially a digital scratch pad that may be used to compare up to seven discrete abort solutions. One solution is inserted for each manual AST request. These computations are more precise than the tradeoff and less precise than return-to-Earth digital computations. They consist of an impulsive velocity change, an integrated coast trajectory, and polynomial reentry functions. The workflow to generate a Maneuver PAD or target load for a maneuver always involves first an AST calculation followed by using the Return-to-Earth Digitals.

Three types of AST solution can be generated. Unspecified area (time or fuel critical), specific area (ATP or PTP) and lunar search (specific site or fuel critical). The first two of these types work in both Earth and Moon sphere of influence, while the lunar search logic can only be used while in lunar orbit. The calculations that are different between Earth and Moon centered state vectors is chosen internally.

The two nominally used modes are lunar search for TEI and the fuel critical, unspecified area (FCUA) mode for transearth midcourse corrections.

Inputs

TYP: Cycle between the three AST types.

SIT: Choose the landing site or type of abort.

VTI: Choose the vector time for the abort (MPT mode only)

TIM: Choose the time of abort, or minimum abort time in the case of lunar search.

TDV: Choose the maximum DV for a time critical abort or the maximum abort time for the lunar search option.

TZ: Choose the estimated landing time for PTP and ATP aborts.

AST: Go to the AST display page.

ENT: Choose the entry profile.

MD: Choose the maximum miss distance for a PTP abort.

INC: Choose the desired return inclination (lunar reference only). Using 0 as input will optimize the DV.

BCK: Back to entry targeting menu.

Display Buttons

DEL: Delete one or all AST rows.

CLC: Calculate AST solution.

RTE: Go to RTE Digitals inputs page.

BCK: Go back to AST inputs page.

Display Explanation

CODE	SITE AM	GETI GETV	DV INCL HPC	VEI GEI	GETEI GETL	LAT IP LNG IP
101	MPL LSATP	095:59:38 094:01:50	2927 A40.00 58.2	36126 -6.50	169:40:37 169:48:59	03:09N 167:06W

Figure 8: Example Abort Scan Table

CODE: Code associated with the AST solution. Starts at 101 and is incremented with each new calculation.

SITE: When using PTP or ATP mode the specified landing site is shown, e.g. MPL for Mid Pacific Line.

AM: Abort Mode. First letter is E for Earth or L for Lunar centered. Second letter is S for lunar search or D for discrete time. Remaining letters show the abort type, ATP, PTP, TCUA or FCUA.

GETI: GET of the impulsive ignition.

GETV: GET of the state vector time used in the calculation.

DV: Total Delta V of the maneuver.

INCL: Earth relative inclination of the trajectory at entry interface. A for ascending and D for descending in terms of azimuth.

HPC: Height of pericynthion, only calculated in lunar reference.

VEI: Velocity at entry interface.

GEI: Flight path angle at entry interface.

GETEI: GET of entry interface.

GETL: GET of splashdown.

LAT IP: Latitude of impact.

LNG IP: Longitude of impact.

3.9.4 Return-to-Earth Digitals

The final solution produced by the RTE section — the Return-to-Earth digital solution — consists of an integrated maneuver, an integrated trajectory from the end of the maneuver to entry interface, and an integrated reentry. Two solutions, each the result of a manual request, may be viewed with the Return-to-Earth digital display. Either of these solutions may be transferred into the Mission Plan Table, used to initiate execution of the spacecraft setting study aid, or used to generate a command load.

Computation of a solution is similar to the AST computations in that the same iteration algorithm is used to adjust independent and dependent variables to meet certain constraints on the dependent variables. The major differences can be listed:

1. The first guess is a converged solution being viewed in the AST Display.
2. The constraints at entry interface are taken from the entry interface state of the converged solution.
3. Independent variables are the target parameters for the finite maneuver integrator (PMMRKJ).
4. The reentry parameters are always obtained from an integrated reentry after the iteration converges to the entry interface conditions.

The same precision trajectory logic is used by the iteration algorithm except that now the finite maneuver integrator is used to perform the maneuver instead of making an impulsive velocity change. The user may specify to the Return-to-Earth digital computations any of four thrusters (SPS, SMRCS, DPS, or LMRCS). In addition to requesting a solution constructed as described above, the user may manually define a solution by defining, via the manual entry device (MED) a time for a vector fetch, a time of abort, and maneuver targets. This type of manual entry bypasses the iteration logic. The coast Encke integrator

is used to propagate the fetched vector to the time of abort. The maneuver is integrated one time using the targets supplied. The coast Encke is used again to propagate the burnout vector to 400,000 feet at which point the reentry integrator is used to propagate to impact.

Inputs

COL: Solution will be shown in either primary or manual column.

AST: Choose AST code for the calculation.

REF: Choose REFSMMAT type for the calculation. This can be any REFSMMAT currently stored in the RTCC or any of the following additional types that are calculated with the RTED solution:

- **ROP:** Attitude is 0, 0, 0 in burn attitude with heads up. This is identical to the preferred REFSMMAT in the AGC.
- **ROY:** Attitude is 180, 180, 0 in burn attitude with heads up. This is the Earth orbit retrofire REFSMMAT.
- **ROZ:** Attitude is 0, 180, 0 in burn attitude with heads up.
- **TEI:** Attitude is 180, 0, 0 in burn attitude with heads down. This is the TEI REFSMMAT used on Apollo 12 and later.
- **DEI:** Attitude is 0, 180, 0 at reentry in an LVLH orientation with heat shield forward, inplane, lift up, heads down. This is the standard lunar entry REFSMMAT.
- **REI:** Attitude is 0, 0, 0 at reentry in an LVLH orientation with heat shield forward, inplane, lift up, heads down.

MAN: Choose maneuver code for the calculation. The code consists of four letters. The first letter is the spacecraft performing the maneuver, C for CSM and L for LM. The second letter is the thruster used for the maneuver. S for SPS, D for DPS or R for RCS. The third letter is the spacecraft configuration. D for docked, A for ascent stage docked and U for undocked. The last letter is always X for External DV guidance.

ULL: Choose the ullage thrusters and duration for the burn. The ullage thruster options are 2 or 4. If the burn is an RCS burn then the options are +2, +4, -2 or -4 and these will be the thrusters used for the maneuver.

TRM: Choose the trim angle option (calculated or system parameter)

DIS: Go to the RTE Digitals display

DOC: Choose the docking angle during the maneuver.

HEA: Choose heads up or down for the maneuver.

ITE: Choose iterate or not iterate for the maneuver. Iterate is the more accurate solution, but takes longer to calculate and has a small risk of not converging.

BCK: Go back to entry targeting page.

Display Buttons

CLC: Calculate RTE digitals solution.

SPL: Save the splashdown target from either primary or manual column.

TRA: In non-MPT mode the TIG and DV are save to be used for uplink and Maneuver

PAD. In MPT mode the maneuver gets transferred to the MPT. The input is the MED format M74. To transfer the primary column enter "M74,CSM,,RTEP;" for the manual column "M74,CSM,,RTEM;"

BCK: Go back to RTE Digitals inputs page.

Display Explanation



Figure 9: Example Return-to-Earth Digitals

GETR: Reference time in GET of an event.

STA ID: Station ID of the state vector used in the calculation (MPT mode only).

AM: Abort mode, see AST display explanation.

GETV: GET of vector, see AST.

AREA: Splashdown area, see AST.

THR: Maneuver code, see RTED inputs.

MATRIX: REFSMMAT used for the calculation.

WT: Total weight at main engine on.

TAA: True anomaly after abort.

EP: Primary entry profile used to generate entry simulation. HGN for G&N or HB1 for constant G reentry.

RLH: Roll angle at ignition in LVLH coordinates.

PLH: Pitch angle at ignition in LVLH coordinates.

YLH: Yaw angle at ignition in LVLH coordinates.

RO: Roll/outer gimbal angle at ignition in IMU coordinates.

PI: Pitch/inner gimbal angle at ignition in IMU coordinates.

YM: Yaw/middle gimbal angle at ignition in IMU coordinates.

VC: Delta V to be used in the EMS DV counter for the burn.

BT: Burn time of the maneuver, main engine on to cutoff.

VT: Total Delta V of the maneuver.

U: Number and direction of RCS thrusters used for ullage or as the main engines for the maneuver.

DT: Ullage duration.

PETI: Phase elapsed time of ignition, relative to GETR.

GETI: Ground elapsed time of ignition.
GMTI: GMT (since midnight launch day) of ignition.
BU: Backup entry profile.
PETIR: Phase elapsed time of initial roll (usually 0.05g)
LV: Lift vector orientation, initial roll angle.
GIR/GCON: G level of initial roll if constant G iteration was the backup entry profile. Otherwise constant G level to be used to generate backup impact coordinates.
GMAX: Maximum G level encountered during the reentry.
PETEI: Phase elapsed time of entry interface.
VEI: Velocity at entry interface.
GEI: Flight path angle at entry interface.
LAT LNG EI: Latitude and longitude at entry interface.
LAT LNG ML2: Latitude and longitude at splashdown if primary entry mode skipped out and maximum lift was used for second entry
LAT LNG T: Latitude and longitude of the splashdown target
LAT LNG ZL2 Latitude and longitude at splashdown if primary entry mode skipped out and zero lift (ballistic) was used for second entry
LAT LNG IPB: Latitude and longitude at splashdown with the backup entry mode.
GETL: GET at drogue chute deployment using the primary entry mode:
MD: Miss distance of the primary entry mode to the target splashdown coordinates. In nautical miles.

3.10 Skylab Launch Window Processor

The launch window processor (LWP) can be used to calculate the optimal liftoff time and launch targeting parameters for Saturn IB missions to a rendezvous target in orbit, such as Skylab and ASTP. In the input descriptions below GMTLO* refers to the lift-off time that is required to achieve zero phase angle to the target at insertion.

3.10.1 Inputs

TGT: Select target vehicle in orbit for rendezvous.

LOT: Liftoff time option. The options are:

- "Input time", lift-off on input time.
- "Phase angle offset", compute lift-off time to achieve a desired phase angle (OFFSET) at insertion.
- "Biased phase zero (GMTLOR)", lift-off on GMTLO* plus BIAS, using input time as threshold.
- "Biased phase zero (TPLANE)", lift-off on GMTLO* + BIAS, using TPLANE as threshold time.
- "In-plane", lift-off based on inplane launch time ($\text{GMTLO} = \text{TPLANE} + \text{TRANS}$).
- "In-plane with nodal regression", iterate on lift-off on inplane launch time based on target orbit phase angle (final $\text{GMTLO} = \text{TYAW} + \text{TRANS}$).

CKFACT: Chaser vehicle K-factor (drag multiplier). Not used yet.

CAREA: Chaser vehicle reference area. Not used yet.

CWHT: Chaser vehicle weight at insertion. Not used yet.

NS: Compute a launch using the northerly or southerly launch opportunity.

DAY: Day on which launch times are computed, relative to base date.

PFT: Powered flight time.

PFA: Powered flight arc.

YSMAX: Yaw steering limit.

DTOPT: Delta time to be subtracted from analytical inplane launch time to obtain empirical inplane launch time.

DTGRR: Delta time from lift-off, which defines the time of guidance reference release.

RINS: Radius of insertion in meters.

VINS: Velocity of insertion in meters per second.

GAMINS: Flight-path angle of insertion in degrees.

GMTLOR: Threshold or actual lift-off time for LOT options that require an input time.

OFFSET: Phase angle desired at insertion.

BIAS: Bias that is added to GMTLO* to produce lift-off time.

TRANS: Delta time added to inplane time to obtain lift-off time.

INSCO: Insertion cutoff conditions options:

- "Input VINS, GAMINS, RINS", use the input parameters for insertion.
- "Input GAMINS, RINS and height difference", calculate the insertion velocity to achieve the height difference DHW to the target at apogee after insertion.

- "Input GAMINS, RINS, altitude at angle from insertion", calculate the insertion velocity to achieve the altitude DHW at the angle DU after insertion.

DHW: Desired height difference between chaser and target, or altitude of chaser at input angle from insertion.

DU: Angle from insertion to obtain a given altitude or delta altitude.

ANOM: Nominal semimajor axis at insertion, used for an initial guess of the insertion velocity in the INSCO options that calculate it.

DELNOF: Flag for option to compute differential nodal regression from insertion to rendezvous.

DELNO: Angle that is added to the target descending node to account for differential nodal regression.

PHASE: Expected phase angle at insertion. The default option is $90\text{-}270^\circ$, which should work for most of the $0\text{-}360^\circ$ range. Near 360° some difficulty to converge could arise, so if a calculation fails then switching this option to $270\text{-}540^\circ$ can help. Other than that there should be no need to change the default value.

DIS: Go to launch targeting display.

BCK: Go back to utilities page.

3.10.2 Display



Figure 10: Launch Targeting Display for Skylab 2

GMTLO: Greenwich mean time of lift-off, in hours, minutes, seconds

TINS: Greenwich mean time of insertion, in hours, minutes, seconds

GMTLO*: Greenwich mean time of phase match, in hours, minutes, seconds

PFA: Powered flight arc.

PFT: Powered flight time.

DN: Descending node of chaser.

TPANE: Greenwich mean time of inplane launch, in hours, minutes, seconds

AZL: Optimum launch azimuth.

LATLS: Geocentric latitude of launch site.

LONGLS: Geographic longitude of launch site.

DELNO: Angle between the target and chaser descending nodes, defined at insertion.

DELNOD: Rate of change of DELNO, defined at insertion.

TYAW: Greenwich mean time of lift-off to achieve minimum yaw steering, in hours, minutes, seconds

TGRR: Greenwich mean time of guidance reference release

VIGM: Velocity magnitude at insertion, in meters per second and feet per second.

RIGM: Radius magnitude at insertion, in meters and feet

GIGM: Flightpath angle at insertion

IIGM: Inclination at insertion

TIGM: Angle measured from launch site meridian to chaser descending node, defined at TGRR

TDIGM: Rate of change of TIGM, defined at TGRR

APOGEE: Height of apogee in nautical miles

PERIGEE: Height of perigee in nautical miles

INCLINATION: Inclination of orbit plane

INS PHASE: Phase angle at insertion
DN TARGET: Descending node of target
BIAS: Time added to GMTLO* time to obtain a lift-off time
T ANOMALY: True anomaly
ALTITUDE: Height of chaser at, insertion
DH: Delta height between chaser and target, at insertion
TIME: Greenwich mean time of lift-off

4 Pre-Advisory Data

4.1 Maneuver PAD

4.1.1 Explanation

The Maneuver Pre-Advisory Data (PAD) contains all necessary numbers to safely conduct a burn with the SPS or RCS. A complete explanation of each item on the PAD can be found in all Apollo flight plans, e.g. [here](#). Additionally to the Maneuver PAD the very similar Apollo 7 TPI PAD was added as a second mode.

4.1.2 Buttons

VEH: The vehicle configuration is only displayed here and chosen on the configuration page.

ENG: Choose between the Service Propulsion System (SPS) and the Reaction Control System (RCS) for the maneuver.

HEA: Choose between conducting the maneuver in a heads-up or a heads-down orientation.

TIG: If you want to display a Maneuver PAD for a maneuver not calculated with the Apollo RTCC MFD you can manually enter the desired Time of Ignition and Delta V.

DV: See above.

CLC: Calculate the missing numbers on the Maneuver PAD.

OPT: Switch between the Maneuver PAD, the Apollo 7 TPI PAD and the TLI PAD.

REQ: Request a maneuver solution calculated with LTMFD or IMFD.

BCK: Go back to the main menu.

4.2 Entry PAD

4.2.1 Explanation

The Entry PAD contains all numbers to conduct a safe reentry. There are two types of Entry PADs: Earth Orbit Reentry and Lunar Entry. A complete explanation of each item on the PAD can be found in most Apollo flight plans, e.g. [here](#).

4.2.2 Buttons

MAN: For a lunar entry you can choose between calculating a direct Entry PAD without any additional maneuvers or a Entry PAD for a previously calculated Midcourse Correction. For an Earth orbit entry a deorbit maneuver has to be performed in any case.

DWN: Downlink the splashdown coordinates from the AGC.

CLC: Calculate the missing numbers on the Entry PAD.

OPT: Switch between the Earth Entry PAD and the Lunar Entry PAD.

BCK: Go back to the main menu.

4.3 Landmark Tracking

4.3.1 Explanation

On the Landmark Tracking page coordinates on the spherical bodies (Earth and Moon) in Orbiter 2010 can be converted to AGC coordinates. Also the contents of a Landmark Tracking PAD can be calculated. These are used for the correct timing of a pitchdown maneuver for better tracking with Program 22.

T1 is the time at which the CSM comes over the horizon and becomes visible from the landmark. At this time the astronaut can begin looking at the landmark to find the specific point he wants to track.

T2 is the time at which the CSM is at an elevation angle of 35° from the landmark. If any marks on the landmark are to be done, then at this time the pitchdown maneuver should be started. In Earth orbit this is usually $0.5^\circ/\text{s}$, in lunar orbit $0.3^\circ/\text{s}$.

The other displayed values are the distance of the landmark from the ground track of the CSM orbit and the AGC inputs. The AGC uses geodetic latitude, longitude divided by 2 and altitude in nautical miles as the inputs.

4.3.2 Buttons

TIM: Estimated time over the landmark.

LAT: Geocentric latitude of the landmark. If the landmark is listed in a marker file, then that latitude should be used as an input here.

LNG: Longitude of the landmark.

CLC: Calculate AGC coordinates and Landmark Tracking PAD.

4.4 Map Update

4.4.1 Explanation

The Map Update is very different in Earth and Moon orbit. In Earth orbit the next ground station with the times of acquisition and loss of signal (AOS and LOS) are displayed. In lunar orbit a few more times are displayed: loss of signal (LOS), sunrise (SR), crossing of the prime meridian (PM), acquisition of signal (AOS) and sunset (SS) are shown. These values are written down on the Apollo 8 Map Update forms.

4.4.2 Buttons

CLC: Calculate map update.

MOD: Cycle between Earth and Moon orbit.

4.5 AGS State Vector Update

4.5.1 Explanation

This page shows the AGS State Vector Update PAD. There would also be a MOCR display for the same function (MSK 277) but this page has been modelled after the PAD given to the crew for convenience. The AGS SV PAD can be used to manually input CSM and LM state vectors on the DEDA. This is a backup method, the main method of updating the state vectors would be via downlink from the PGNS (Verb 47).

Because the AEA uses no REFSMMAT the coordinate system of the state vectors depends on the current alignment being used. For that the current REFSMMAT in use can be selected. The RTCC keeps several REFSMMAT stored, the main ones for this page would be the current (CUR) or AGS REFSMMATs. As the PGNS and AGS typically use the same alignment the CUR REFSMMAT would be saved as the AGS REFSMMAT and then the AGS REFSMMAT is used for the state vector calculation. But any REFSMMAT in the LM REFSMMAT table can be used.

This page can also be used to manually or automatically update the AGS K-Factor. For the automatic option the current PGNS and AGS clocks are compared and the difference is then saved as the K-Factor.

4.5.2 Buttons

SLT: Cycle between updating the CSM or the LM part of the PAD.

TGT: Update the CSM or LM vessel.

REF: Choose the REFSMMAT in the LM REFSMMAT table to use for the state vector calculation.

AGS: Manually enter the AGS K-Factor.

KFA: Run the automatic calculation to update the K-Factor.

CLC: Calculate the AGS state vector PAD. CSM and LM state vectors are calculated separately.

SAV: Save the present LM CUR REFSMMAT as the AGS REFSMMAT. All REFSMMATs in the LM REFSMMAT table can be used for the state vector calculation, but only the AGS REFSMMAT is used in converting telemetry vectors (see Vector Panel Summary) so this is a simple way to update the AGS REFSMMAT. The alternative would be using the G00 MED.

5 Utilities

5.1 REFSMMAT

5.1.1 Explanation

The REFSMMAT (REference to Stable Member MATrix) is a rotation matrix relating the Apollo Basic Reference Coordinate System (BRCS) and the currently used IMU Stable Member Coordinate System. Depending on the mission phase the REFSMMAT is chosen, so that the IMU angles provide meaningful attitude values. Some types of REFSMMATs can be calculated by the AGC itself, but most were uplinked to the spacecraft from the ground. The REFSMMATs that can be calculated with this MFD are:

- Launch: Calculates the Launch REFSMMAT, which is also calculated internally in the AGC at liftoff.
- Landing Site: Not used for Apollo 7 or 8
- PTC: Passive Thermal Control, not used for Apollo 7 or 8.
- LOI-2: A special LVLH REFSMMAT for Apollo 8, calculated before the last translunar Midcourse Correction.
- P30: Alignment for a thrusting maneuver, equivalent to option 1 in Program 52.
- P30 retro: Alignment for a retrograde burn, useful for Earth orbit reentry maneuvers.
- LVLH: Local Vertical alignment, equivalent to option 2 in Program 52.
- Lunar Entry: Equivalent to option 2 in Program 52 with the GET of Entry Interface.

5.1.2 Buttons

TIM: The options "Landing Site", "PTC", "P30", "P30 retro " and "LVLH" require a time in GET to calculate the REFSMMAT. For a Landing Site REFSMMAT the time chosen is either the predicted landing or launch time. The time for P30 and P30 retro REFSMMATs is the maneuver time and is set on a maneuver calculation page (Lambert, CDH or Entry).

TYP: Choose between uplinking the REFSMMAT or the desired REFSMMAT. The desired REFSMMAT is the alignment, that Program 52 will align the platform to, based on the knowledge of the attitude referenced to the old, currently used REFSMMAT. Only in rare cases the REFSMMAT itself would be uploaded, e.g. when activating the Lunar Module or if the difference to the previous REFSMMAT is very small. In doubt, uplink the desired REFSMMAT!

DWN: Downlink the current REFSMMAT from the AGC. If the type of REFSMMAT is known, select it by cycling through the REFSMMAT types by pressing OPT before doing the downlink. Useful for calculating PADs with a REFSMMAT not calculated by the RTCC MFD.

MCC: The calculated REFSMMAT usually depends heavily on the current orbit. If there is a maneuver between now and the set time or the reentry time, change the setting to MCC to take the maneuver into account. The LOI-2 REFSMMAT is special, because the calculation of two maneuver is required before the LOI-2 REFSMMAT can be calculated. This will be explained in more detail on the Lunar Insertion page.

OPT: Switch between the different options.

CLC: Calculate the REFSMMAT.

UPL: Uplink the REFSMMAT to the AGC.

LAT: Only for Landing Site: Choose the latitude of the landing site.

LNG: Only for Landing Site: Choose the longitude of the landing site.

BCK: Go back to the main menu.

5.2 Apollo Generalized Optics Program

The Apollo Generalized Optics Program (AGOP) contains various optics and antenna pointing calculations.

Option 1 of the AGOP will be used for cislunar navigation. This program option defines an inertial attitude which will align the optics system to the horizon or some specified landmark on the Earth or Moon. Output consists of IMU gimbal angles and the optics shaft and trunnion angles to point the sextant at the specified star.

Option 2 will accept a state vector and compute the right ascension and declination of the spacecraft with respect to the Earth, and the right ascension and declination of the Earth, Moon and Sun with respect to the Earth. Additionally it computes the same quantities referenced to an input landmark when mode 2 is selected. Mode 2 will compute right ascension, declination, and a unit vector from the spacecraft to the specified landmark or the center of the Earth, Moon and Sun as desired

Option 3 will output the right ascension, declination and unit vector of all stars in the catalog.

Option 4 will accept a state vector and IMU gimbal angles and compute the pitch and yaw angles of the onboard S-Band Hi-Gain Antenna, S-Band Steerable Antenna, and Rendezvous Radar Antenna necessary to point at a specified ground based radar. An additional option provides the capability of fixing the antenna position and computing the necessary spacecraft gimbal angles for pointing the antenna at a specified ground station.

Option 5 computes the IMU gimbal angles required to place the spacecraft in a passive thermal control (PTC) attitude, with the +X-axis oriented 90 degrees with respect to the Sun and to the Earth for omnidirectional communication.

Option 6 computes the attitude required to orientate the CSM to the forward or aft horizons, either in heads up or down.

Option 7 (Optics Support Table) will primarily be used to verify the CMC and LGC stable member alignment made by using the onboard optical sighting equipment. This option is divided into 5 modes which are as follows:

Mode 1 computes the yaw angle required to place the LM z-axis in the local vertical plane and a pitch angle which will place the horizon on the landing point designator (LPD). The necessary inputs are REFSMMAT, gimbal angles, and time.

Mode 2 uses an input REFSMMAT, spacecraft attitude, and time interval to compute AOS, LOS, and optics angles for 10 stars.

Mode 3 computes a REFSMMAT by specifying star IDs, optics angles, and spacecraft attitude. The capability exists to input two sets of gimbal angles with the above input data and compute the REFSMMAT.

Mode 4 uses an input REFSMMAT and spacecraft attitude for both vehicles to compute the second vehicle REFSMMAT. If both REFSMMATs are input, the second vehicle attitude will be computed.

Mode 5 computes the CSM gimbal angles required to point the AOT at the desired target. The inputs are REFSMMAT, docking angle, one star ID, and optics angles.

Mode 6 computes the attitude for the preferred REFSMMAT. Given a current and a desired REFSMMAT, it computes the gimbal angles for the current REFSMMAT, which define 0, 0, 0 gimbal angles for the preferred REFSMMAT and outputs a set of gimbal angles and FDAI angles which used in conjunction with the current REFSMMAT defines the spacecraft attitude necessary to switch to the preferred REFSMMAT and read 0, 0, 0 gimbal angles.

Options 8 and 9 will be implemented in the future.

5.3 VECPOINT

5.3.1 Explanation

The VECPOINT page, named after a routine in the AGC, is calculating the IMU angles to point a specific part of the CSM or LM in the direction of a celestial object/astronomical body. Any body present in Orbiter 2010 can be chosen. The X-axis of the spacecraft is along its longitudinal axis, so +X is pointing the CSM directly at the body and the SPS engine directly away from it.

5.3.2 Buttons

BOD: Type the name of the body e.g. Sun, Moon etc.

DIR: Choose the direction of the spacecraft to be pointed at the celestial object.

CLC: Calculate the IMU angles.

5.4 S-IVB Lunar Impact Targeting

The Lunar Targeting Program (LUNTAR) was a software available at the Marshall Space Flight Center to calculate the burns of the S-IVB Auxiliary Propulsion System (APS) to cause the S-IVB to impact the Moon in a desired place (Apollo 13 and later). The S-IVB performed a short APS burn and a LOX dump shortly after the CSM/LM separated from it which caused the S-IVB to be on an impact trajectory, but not yet to the desired coordinates. There were usually two midcourse corrections scheduled for the S-IVB, the first one being non-zero.

The targeting available in the RTCC MFD has three different modes. First a Trajectory Evaluation mode which checks where (if at all) the S-IVB will impact the Moon without any further APS burn. Then a Burn Evaluation mode which shows the impact coordinates given an input APS burn. And lastly the actual targeting mode which calculates the required burn to reach the required coordinates. The desired coordinates and the initial guess for the APS MCC-1 can be found in lunar impact postflight evaluation reports which can be found on NTRS, e.g. this one for Apollo 13.

The uplink to the IU for the impact targeting can currently not be done in the RTCC MFD, but in the Project Apollo MFD instead. This might be moved in the future.

5.4.1 Inputs

MOD: Cycle between the three calculation modes described above.

TIG: Enter the time of ignition in Ground Elapsed Time.

BT: Enter the estimated burn time in seconds.

PIT: Enter the estimated LVLH pitch attitude of the maneuver in degrees.

YAW: Enter the estimated LVLH yaw attitude of the maneuver in degrees.

LAT: Enter the desired lunar impact latitude in degrees.

LNG: Enter the desired lunar impact longitude in degrees.

OPT: Either optimize the Delta V of the burn or enforce a specific lunar impact time.

IMP: Enter the desired lunar impact time in Ground Elapsed Time.

5.4.2 Outputs

- Time of ignition referenced to the start of Timebase 8
- Calculated (or input) burn time
- Calculated (or input) LVLH pitch angle in degrees
- Calculated (or input) LVLH yaw angle in degrees
- Actual lunar impact latitude in degrees
- Actual lunar impact longitude in degrees
- Actual lunar impact flight-path angle in degrees

6 Uplinks

6.1 State Vector

6.1.1 Explanation

The state vector of the vessel can be calculated and uplinked here. Additionally to the functionality in the Project Apollo MFD, this MFD can calculate a state vector in the future, which sometimes was used during the Apollo program to prevent an internal state vector integration of the AGC.

The AGC has two slots for state vectors: CSM and LM. For the CSM the MFD will prevent uplinking a state vector that is not the vessel itself. The vessel for the LM can be freely chosen.

6.1.2 Buttons

MOD: Choose between calculating the state vector "now" and at a specified GET.

TIM: Set the desired GET for the state vector in GET mode.

TGT: Set the target vessel.

SLT: Switch between the slots.

CLC: Calculate a state vector.

UPL: Uplinks the calculated data to the AGC.

BCK: Go back to the main menu.

7 Mission Planning

The mission plan table (MPT) is the central planning tool in the RTCC for future maneuvers. The RTCC has two MPTs, one for the CSM and one for the LM, which are related through undocking and docking maneuvers. Each MPT consists of a table header and data blocks for each individual maneuver. The table header has the initial vehicle configuration (combinations of CSM, Saturn and LM ascent and/or descent stages) and initial weights, area (for drag calculations) and propellant. Each maneuver block has data describing the maneuver and updates weights and propellant.

At the current stage of RTCC development for NASSP the MPT is an optional feature that can be used to plan and evaluate future maneuvers. The RTCC MFD has a MPT mode which by default is off. When it is switched on then current state vectors and weights are not updated automatically but have to be managed manually. However, the MPT header is already used in the non-MPT mode for trajectory propagation for state vector uplinks. In the future more parts of the MFD will use the data in the MPT header.

There are four manual user entry (MED) modes to update the MPT header. The M55 MED is used to establish the initial vehicle configuration, time to start S-IVB venting and the initial CSM/LM docking angle. M51 is used to update the vehicle areas and a K-factor (multiplier) for drag calculations. In MPT mode the M50 MED is used to update vehicle weights and M49 is used to update available propellants. The detailed MED inputs and format can be found in the MED section of the manual.

7.1 MPT Initialization

On the left side of this page appear the input parameters that are automatically or manually updated. On the right side the page shows the actual data stored in the MPT.

7.1.1 Buttons

TAB: Choose the mission plan table (CSM or LM) to initialize.

MED: Choose the type of initialization to do. The options are M49 for propellants, M50 for weights, M51 for vehicle areas or M55 for the initial vehicle configuration.

UPD: To update the MPT (right side) with the data currently set as input (left side).

<<: Go to previous item to manually change input data.

>>: Go to next item to manually change input data.

SET: Change the currently selected input.

TGT: Select a vessel in the simulation to automatically populate the input data.

AUT: Automatically update the input data with the selected vessel.

VPS: Go to the Vector Panel Summary page for MPT state vector management.

BCK: Go back to the main menu.

8 Configuration

MIS: Leads to a page where several files can be selected that load data into the RTCC. The system parameters file contains e.g. launchpad coordinate, AGC uplink addresses. These numbers will be constant for the mission, no matter on what launch day the mission is launching. The TLI file contains parameters for simulating the TLI maneuver in the RTCC. This is required for the TLI PAD and adding a TLI maneuver to the MPT. The file contains data for whole monthly launch window. The Skeleton Flight Plan (SFP) file contains targets and initial guess for the TLI and MCC processors. They are also for the whole monthly launch window.

TYP: Choose the type of vehicle configuration (CSM or LM, docked or undocked).

STA: Choose the type of LM configuration that is currently being used, ascent or descent configuration.

SXT: Change the time of the sextant star check, which is part of the procedure for a normal maneuver. During Earth orbit missions the Earth often blocks the sextant from viewing many stars, so adjusting the time of the check before the maneuver allows the MFD to find a suitable star.

UPL: Inhibit or enable uplinks during times of no available ground stations. Currently all ground stations being used for Apollo 7 are implemented.

DAT: Choose the launch date for the mission.

TIM: Choose the launch time for the mission. This will update the launch time of the CSM stored in the RTCC.

EPO: Choose the AGC epoch. Usually this is a MJD at around January 1st of the yearly coordinate system defining period. This value should be automatically chose correctly for the AGC version in use.

UPD: Update the liftoff time automatically by downlinking that time from the CMC or LGC. This will actually update three values in the RTCC: CSM liftoff time, time of clock zeroing in the CMC, time of clock zeroing in the LGC. These times are normally all set to the actual liftoff time of the CSM to get a consistent basis to calculate Ground Elapsed Time in the RTCC.

BCK: Go back to the main menu.

9 Example: Apollo 7 Rendezvous

This MFD can be used to replicate the ground solutions for the rendezvous and other SPS burns during the Apollo 7 mission. As an example the inputs for the following maneuvers will be presented:

1. NCC1 burn at 26:25:00 GET.
2. NCC2 burn at 27:30:00 GET.
3. NSR burn at 28:01:00 GET.
4. TPI burn at ca. 29:25:00 GET.

9.1 NCC1 burn

At 26:25:00 GET a SPS burn was executed that will put the CSM on a trajectory resulting in a phase angle of 1.32° behind and 8NM below the SIVB at 28:00:00GET. The maneuver is calculated with the Two-Impulse Multiple Solution display. The required inputs are:

- VEH is set to CSM.
- Both maneuver times are fixed (IV option)
- The Chaser vessel is AS-205, the target vessel is AS-205-S4BSTG.
- T1 is set to 26:25:00 GET (NCC1 time).
- T2 is set to 28:01:00 GET (NSR time).
- The desired offsets are 8.0 NM for DEL H, 1.32° for PHASE, 27.45° for ELEV (at TPI) and 140° for WT (travel angle from TPI to TPF)

On the TI Multiple Solution display only this one maneuver will appear. With the TRA button the maneuver can be converted to a finite burn, taking the maneuver engine into account (SPS). The resulting burn solution should be close to the vector (+62.7, -1.0, +154.8). This can be used in a P30/P40 automatic SPS burn with the CSM.

9.2 NSR burn

The NSR burn nominally happens at 28:00:00 GET and places the CSM in a coelliptic orbit with a constant delta height to the target. On the CDH page of the MFD the inputs for the burn are the GET (028:00:00) and the Delta Height (DH) of the orbit, which is 8 NM for Apollo 7. Because the GET is variable, the option "Find GETI" should be used. A positive value here means below the target. When calculating the burn, the new time for the maneuver is also displayed below the number for DH. The new time is chosen, so that the delta height of the burn is exactly the specified 8NM. The results should be close to:

- 028:00:30 GET
- DX: -92.7 fps
- DY: +1.6 fps
- DZ: -106.2 fps

These numbers can be used for the external DV program (P30).

9.3 TPI burn

The TPI maneuver nominally was calculated by the AGC itself, but a backup solution was calculated on the ground. This backup solution can be replicated with the MFD.

On the Two Impulse Multiple Solution page first set the S-IVB as the target. Then press T1 and enter any negative number. The MFD will now try to find the T1, when an elevation angle of 27.45° occurs, provided this angle has been entered using the P51 button. To find the T2 time, which is 140 degrees of orbital travel after T1, press T2 and enter a negative time again. T2 will now be calculated in the TI processor. Usual values for the maneuver:

- 29:21:38 GET
- DX: +13.7 fps
- DY: +0.9 fps
- DZ: -7.9 fps

On the Maneuver PAD page press OPT and CLC to display the TPI PAD.

10 Example: Midcourse Correction Planning

10.1 Example 1: Apollo 11 MCC-2

As mentioned in the introduction of this section, the version of the midcourse correction processor implemented in the RTCC MFD was used for Apollo 14 to 17. Apollo 11 would have used mode 2 of the processor for their MCC-2. Modes 2 and 4 were changed to only apply to the LOI/DOI maneuver sequence of those later missions. The same capability was retained in modes 3 and 5 though, by inputting the same desired azimuth as min and max azimuth on the constraints page.

To start off the calculation, go to the SFP page, under Maneuver Targeting, Translunar, Skeleton Flight Plan Table. The SFP contains initial guesses and some targeting parameters that the MCC calculation requires. The premission data is stored in tables that are valid for the whole daily launch window, so the first step is to select the data set for the actual launch conditions. Press the F62 button and check that the first TLI opportunity and the correct launch azimuth usually 72 deg) are entered. After pressing enter the MFD page should now be populated with data. This step has to be done just once per mission and otherwise no changes will have to be done on the SFP page.

The next step happens under Maneuver Targeting, Translunar, Midcourse Processor. This takes you to the computation page of the processor. Here choose mode 3 (click twice on the MAN button). The Apollo 11 MCC-2 happened at about 26:45:00 GET, so press the TIG button and input that time. The other inputs can be left as they are. The solution will be shown in column 1, the maneuver is docked and the skeleton flight plan table no. 1 contains the preflight estimates. Press CON to check that all the constraints are as desired, especially the min and max azimuth constraints being identical. This should already be preloaded in the MFD, so no changes are necessary.

Back to the previous page (BCK button) and then to the midcourse tradeoff display (MID button) and everything should be ready for the calculation. Press CLC and the solution for the mode 3 calculation should be displayed in column 1. Using the Apollo 11 Before MCC-2 scenario that comes with NASSP this results in a maneuver of 19.7 ft/s (DV MCC).

You can now try different inputs and constraints, but if you are happy with the solution, you should now save the resulting data table from the MCC-2 calculation for use in the later MCC-3 and MCC-4 calculations. That is done by pressing the F30 button and typing: *F30,1*; The result can be checked under MCC Display, MSK button, "1597" input, F31 button. The F31 cycles between the preflight (table 1) and the nominal (table 2) targets. Only table 2 will be saved in scenarios.

Back to the Midcourse Tradeoff page, the maneuver still has to be converted from an impulsive, instant maneuver to a finite maneuver taking the thruster being used into account. For that click on the MPT button and then on the THR button until the thruster of your choice is selected. SPS is set by default and is the correct choice for the maneuver. Click the CLC button and the actual TIG and DV have now been generated. These can be used to display e.g. a Maneuver PAD for the midcourse.

10.2 Example 2: Apollo 11 MCC-4

MCC-4 will use the nodal targets (latitude, longitude, radius and time of the desired position at LOI, if it was an instant velocity change maneuver) that resulted from the MCC-2 calculation, and were stored in skeleton flight plan table number 2. MCC-4 is a mode 1 maneuver with a time of ignition of about 70:55:00 GET. Input this value with the TIG button, then press the SFP button so that it says 2, for SFP table 2. Go to the midcourse tradeoff display and press CLC. Converting it to a finite maneuver works the same way as for MCC-2. Possibly MCC-4 will be small enough to be done with the RCS.

10.3 Example 3: Apollo 12 MCC-2

Apollo 12 was the first mission to fly the so called hybrid mission profile. After TLI the trajectory is ideally free return, but with a pericynthion altitude higher than necessary for successful lunar orbit insertion. Therefore there has to be a midcourse correction that takes the trajectory to a close encounter with the Moon, but in the process making it non-free return. This maneuver was planned for MCC-2. After MCC-2 the same LOI-1 and LOI-2 maneuvers were flown as on the previous missions.

To accomplish the hybrid transfer maneuver, mode 5 of the midcourse processor has to be used. Use a TIG of about 30:53h GET, column 1, docked vehicle configuration and table 1 (preflight targets) of the skeleton flight plan. Use the MID button to go to the midcourse tradeoff display and then press CLC to calculate the solution. It should now have calculated fairly large maneuver, the nominal Delta V being 68.8 ft/s.

The trajectory after MCC-2 is not constrained to be free return, but was instead optimized for the smallest possible Delta V. This will also potentially have moved the time of ignition for LOI-1 away from the flight plan time. If it is not critical that the DV optimal solution for MCC-2 is being used, the LOI-1 TIG can be adjusted by constraining that time on the constraints page. This was done on the actual Apollo 12 mission and is not possible for free-return missions like Apollo 11.

The flight plan time for LOI-1 is 83:25:18.2 GET. Check which time ("GET LOI") the midcourse tradeoff page is showing after calculating the MCC-2 solution. If the times are different by more than a few seconds go to the midcourse constraints page and press the F23 button. The times that are input here are not directly the LOI GET, but instead the pericynthion time, which is a bit later than LOI-1. The convergence of the constrained solution works best if the minimum and maximum times input with the F23 button are 10 minutes apart. Usually the LOI TIG will now run into the lower or upper end of the constraint. As you can't predict on which end of the time window LOI will now be, this process is always one of trial and error. So start by inputting a min and max time that contains the GET of LOI. Then recalculate the solution on the midcourse tradeoff page and check what the new LOI GET is. By moving the time window to later or earlier times with the F23 button the LOI GET can be moved as well. This might take a few iterations until the LOI GET is close to the desired time. A few seconds off are acceptable

After this solution resulting in the desired LOI-1 time has been calculated the process is the same as for the Apollo 11 MCC-2. After committing to performing the maneuver as calculated use the F30 button to store the data table containing the numbers required to

calculate MCC-3 and MCC-4. These are calculated with mode 1 just like for Apollo 11.

10.4 Example 4: Apollo 13 MCC-2

11 Example: Deorbit Targeting

11.1 Example 1: Apollo 11 Pre TLI Abort

Should the mission be aborted at the earliest time after reaching orbit, the first deorbit opportunity is the Atlantic ocean area, just one orbit after launch.

The CSM/S-IVB separation procedure used in computing this reentry is a 5-second SM RCS burn in the retrograde horizon monitor attitude. The separation is begun 20 minutes prior to the deorbit burn and places the CSM below and behind the S-IVB at retrofire. The horizon monitor attitude mentioned here is the nominal deorbit attitude and is attained by aligning a mark on the CM window with the earth horizon.

From the circular 100 NM orbit the velocity vs. flight path angle targeting for entry interface cannot be used, as there would be too little time between cutoff of the retrofire maneuver and reentry. Therefore a fixed Delta V of 325 ft/s is used, which leads to a slightly shallow reentry, but enough time for the procedures before reentry.

To start setting up for this calculation, review the MFD page with the inputs for the separation maneuver. The page is located under Maneuver Targeting, Deorbit, Separation/Shaping Constraints. The default values for all inputs are set up for this type of separation maneuver. The LVLH pitch angle corresponds to placing the window mark on the horizon from a 100 NM orbit. So nothing has to be changed on this page.

Next go to the Retrofire Constraints page. Here only the retrofire mode has to be changed from "V, Gamma" to "DV". Then enter a DV of 325 ft/s with the VAL button. The other constraints should be correctly set up. SPS engine, 31.7 deg window line, 4 quads, 15 seconds ullage, CUR REFSMMAT, compute gimbal angles. For the reentry the initial bank angle of 0 deg will be held, followed at 0.2g by a bank angle of 55 deg.

Next the splashdown target is selected on the Target Selection Display. Go to that page, press CLC and enter "1:0:0 -70" for a threshold GET of 1h and the desired splashdown longitude of 70 deg west. The display will now show groundtrack data, starting at that longitude. The time at that longitude should be about 1:40h GET. Now press the SEL button to select the first set of data (at 70 deg west). This will automatically generate the target for the retrofire maneuver.

Now go to the Retrofire Maneuver page. The estimated TIG, latitude and longitude are filled in from the target selection before. The type of maneuver has to be changed to type 2, to include the separation maneuver. Miss distance can be left at 1 NM. Now go to the main output display for the calculation, the Retrofire Digitals, with the DIG button. There the maneuver can be calculated with the CLC button.

11.2 Example 2: Apollo 9 nominal deorbit

The procedure to calculate a nominal deorbit maneuver is very similar to the one described in the previous example. Instead of the fixed Delta V of 325 ft/s the V, gamma target line is used. The calculation is of type 1 (no sep/shaping maneuver), so none of the inputs on the MFD page for the sep/shaping maneuver apply. The splashdown longitude is 59.9 deg west.

11.3 Example 3: Skylab deorbit

The Skylab missions performed the deorbit in two steps. First a SPS burn called the shaping maneuver, which lowers the perigee to 90 NM. Then, about two orbits later, the actual deorbit maneuver, again with the SPS, was done.

The easiest method to target this maneuver sequence is, contrary to the naming, use of the separation burn logic, so that the two maneuvers are separated by a specific time difference. Here the common inputs for the shaping and deorbit maneuver:

Definition of Separation/Shaping Maneuver (R30)

SHA: No Shaping Maneuver

SEP: 180 min (initial guess for two full orbits between the two maneuvers)

THR: SPS

DV: 250 ft/s (initial guess from Skylab orbital altitude)

DT: 0.0 s

ATT: 180.0 -15.0 180.0 (pitched down 15 degrees from local horizontal, heads down)

UDT: 20.0

UTH: 2 jet ullage

GBL: Compute Gimbal Trims

Retrofire Constraints (R31)

ENG: SPS

MOD: V, Gamma

VAL: N/A

ATT: LVLH

LVH: 180.0 -15.0 180.0 (pitched down 15 degrees from local horizontal, heads down)

ULL: 2 quads, 20 seconds

MAT: DES (desired orientation, gives the IMU attitude 0, 180, 0 in burn attitude)

GIM: Compute Gimbal Trims

K1: 0.0 deg

GC: 0.2 gs

K2: 55.0 deg

Retrofire Planning (R32)

TYP: Type 2 (With Sep/Shaping)

Select the target point like in example 1. Two input values have to be tweaked manually with trial and error to get a satisfying solution. First, the Delta V of the shaping maneuver is a manual input, so it has to be changed until the perigee altitude after the maneuver is 90 NM. This value, as well as the other burn parameters of the shaping burn, can be found on the Retrofire Separation (MSK 355) display. In a second step, to optimize the Delta V of the deorbit burn, the time difference from shaping to deorbit has to be adjusted. This is the SEP button on the "Definition of Separation/Shaping Maneuver" page. The deorbit burn should occur near apogee, so tweak this input value until a minimum Delta V of the deorbit burn is reached.

12 Manual Entry Device (MED) Formats

12.1 Acronyms

- **EBCDIC**: Extended Binary Coded Decimal Interchange Code (Characters)
- **FLP**: Floating Point
- **FXP**: Fixed Point (Integer)
- **MSK**: Manual Select Keyboard (display number)

12.2 MED List

MED Code: C10

Purpose: Initiate a CMC/LGC external delta-v update

Example: C10,CMC,1,CSM;

Item Desc.	1	2	3	4	5	6
Item Name	Vehicle Type	Maneuver Number	MPT Indicator			
Input Format	EBCDIC	FXP	EBCDIC			
Input Units						
Checking Option	Exact	Min/Max(2)	Exact(3)			
Missing Item Option	Error(1)	Error	Error			

Notes:(1) CMC, LGC

(2) 1-15

(3) CSM/LEM

MED Code: G00

Purpose: CSM/LM REFSMMAT locker movement

Example: G00,LEM,LLD,CSM,CUR;

Item Desc.	1	2	3	4	5	6
Item Name	CSM/LEM Vehicle	Matrix 1	CSM/LEM Vehicle	Matrix 2	GET	
Input Format	XXX	XXX	XXX	XXX	XXX:XX:XX	
Input Units	EBCDIC	EBCDIC	EBCDIC	EBCDIC	HH:MM:SS	
Checking Option	Exact	Exact	Exact	Exact	≤0Current Time	
Missing Item Option	Error	Error	Error	Error	=Current Time	

Notes: For matrix 1, valid codes are CUR, PCR, TLM, OST, MED, DMT, DOK, LCV, DOD, LLA, LLD, AGS for the LEM and CUR, PCR, TLM, OST, MED, DMT, DOD, LCV for the CSM. For matrix 2, valid codes are CUR, PCR, TLM, MED and LCV for the CSM and CUR, PCR, TLM, MED, LCV, LLA, and AGS for the LEM.

MED Code: G03

Purpose: Compute and save local vertical CSM/LM platform alignment

Example: G03,CSM,100:00:00;

Item Desc.	1	2	3	4	5	6
Item Name	CSM or LEM Vehicle	GET				
Input Format	XXX	XXX:XX:XX				
Input Units	EBCDIC	HH:MM:SS				
Checking Option	Exact					
Missing Item Option	Error	Error				

Notes:

MED Code: M49

Purpose: Change fuel remaining weights

Example: M49,CSM,20000;

Item Desc.	1	2	3	4	5	6
Item Name	MPT Code	SPS Fuel Remaining	CSM RCS Fuel Remaining	S-IVB J-2 Remaining	APS Fuel Remaining	LM RCS Fuel Remaining
Input Format	EBCDIC	FLP	FLP	FLP	FLP	FLP
Input Units	CSM, LEM	lbs.	lbs.	lbs.	lbs.	lbs.
Checking Option		≥ 0	≥ 0	≥ 0	≥ 0	≥ 0
Missing Item Option	Error	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number

∞

Notes: A change of fuel will not cause a trajectory update.

MED Code: M49 (cont.)

Purpose:

Example:

Item Desc.	1	2	3	4	5	6
Item Name	DPS Fuel Remaining					
Input Format	FLP					
Input Units	lbs.					
Checking Option	≥ 0					
Missing Item Option	Insert neg. number					

Notes:

MED Code: M50

Purpose: Change vehicle weights

Example: M50,CSM,40000;

Item Desc.	1	2	3	4	5	6
Item Name	MPT Code	CSM Weight	S-IVB Weight	LM Total Weight	LM Ascent Weight	Time of Weight
Input Format	EBCDIC	FLP	FLP	FLP	FLP	HH:MM:SS
Input Units	CSM, LEM	lbs.	lbs.	lbs.	lbs.	lbs.
Checking Option		≥ 2000	≥ 2000	≥ 2000	≥ 2000	$\leq$ Present Time
Missing Item Option	Error	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number

MED Code: M51

Purpose: Change vehicle cross-sectional area or K-factor

Example: M51,CSM,129.4;

Item Desc.	1	2	3	4	5	6
Item Name	MPT Code	CSM Area	S-IVB Area	LM Ascent Area	LM Descent Area	K-Factor
Input Format	EBCDIC	FLP	FLP	FLP	FLP	FLP
Input Units	CSM, LEM	ft. ²	ft. ²	ft. ²	ft. ²	ft. ²
Checking Option		0 ≤X≤2000	0 ≤X≤2000	0 ≤X≤2000	0 ≤X≤2000	−30 ≤X≤30
Missing Item Option	Error	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number	Insert neg. number

Notes:

MED Code: M55

Purpose: Input initialization parameters

Example: M55,CSM,CSL;

Item Desc.	1	2	3	4	5	6
Item Name	MPT Code	Configuration Code	GET to begin venting	Delta Docking Angle		
Input Format	EBCDIC	EBCDIC	HH:MM:SS	FLP		
Input Units	CSM, LEM	Note 1	GET	degrees		
Checking Option	Exact	Exact				
Missing Item Option	Error	Insert "FF"	Insert neg. number	Insert "FF"		

Notes: Note 1: Any combination of C (CSM), S (SIVB), L (LEM: Ascent+Descent), A (LEM: Ascent Only), D (LEM: Descent Only)

MED Code: P08
Purpose: Update pitch angle from horizon
Example: P08,31.6;

Item Desc.	1	2	3	4	5	6
Item Name	Pitch Angle					
Input Format	FLP					
Input Units	degrees					
Checking Option	None					
Missing Item Option	Error					

Notes:

MED Code: P10

Purpose: Update liftoff time for specified vehicle

Example: P10,CSM,13:32:00;

Item Desc.	1	2	3	4	5	6
Item Name	Vehicle	GMTALO	Traj/No traj Ind.			
Input Format	EBCDIC	H:M:S(.TH)	EBCDIC			
Input Units	VEH	hours	EBCDIC			
Checking Option	Note 1	T $\geq$ 0.	Exact (note 4)			
Missing Item Option	Error	Error	Insert "NO TRAJ"			

- 98 **Notes:**(1) Veh. must be in MHGVNM (CSM, LEM)
(2): First veh. (MGLGMT, MCGMTL), second veh. (MGGGMT, MCGMTS)
(3): Must be "TRAJ" or "NO TRAJ" ("TRAJ" allowed in NPHASE, PRELAUNCH, PRELAUNCH 2 (L.S.))

MED Code: P12

Purpose: Enter GMTGRR and launch azimuth for selected vehicle

Example: P12,CSM,13:32:00,72.0;

Item Desc.	1	2	3	4	5	6
Item Name	Vehicle	GMTGRR	Launch Azimuth			
Input Format	EBCDIC	H:M:S(.TH)	FLP			
Input Units	EBCDIC	hours	deg.			
Checking Option	Exact 1	≥ 0	$70. \leq A \leq 110.$			
Missing Item Option	Error	Error	Error			

Notes: (1) CSM, IU1, IU2. IU1 and IU2 valid at all times. CSM valid only prior to GOST initialization.

(2) See GMSMED documentation (flowchart).

MED Code: P13

Purpose: Enter vector in spherical coordinates

Example: P13,CSM,5521.0,0.0,269.0,0.7,23.4,60.0,102:40:00,L,MCT;

Item Desc.	1	2	3	4	5	6
Item Name	Vehicle	Velocity	Flight Path Angle	Azimuth	Geocentric Latitude	Longitude
Input Format	EBCDIC	FLP	FLP	FLP	FLP	FLP
Input Units	EBCDIC	ft./sec.	degrees	degrees	degrees	degrees
Checking Option	Note 1	≥ 0	$-90.\leq A\leq 90$	$0.\leq A\leq 360$	$-90.\leq A\leq 90$	$-180.\leq A\leq 180$
Missing Item Option	Error					

Notes: (1) Vehicle must be in vehicle name table (CSM, LEM)

MED Code: P13

Purpose: Enter vector in spherical coordinates

Example: P13,CSM,5521.0,0.0,269.0,0.7,23.4,60.0,102:40:00,L,MCT;

Item Desc.	7	8	9	10		
Item Name	Height	Time	Live/Static Ephem. Ind.	Coordinate System Ind.		
Input Format	FLP	H:M:S.TH	EBCDIC	EBCDIC		
Input Units	NM	hours	EBCDIC	EBCDIC		
Checking Option	$H \geq 0$	$T \geq 0$	Exact Note 2	Exact Note 3		
Missing Item Option			Insert L	Insert ECT		

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Notes: (2) Possible inputs:

- L - Live ephem. (default)
- B = put in GZLTRA and STATIC ephem.
- S = STATIC ephem.
- G = Put in GZLTRA only.

(3) Coord. System Ind. = ECT, MCT

MED Code: P15

Purpose: Update GMTZS for specified vehicle

Example: P15,AGC,13:32:00;

Item Desc.	1	2	3	4	5	6
Item Name	Vehicle	GMTZS	DT from GMTZS (LGC) to AGS on			
Input Format	EBCDIC	hours	hours			
Input Units	EBCDIC	hours	hours			
Checking Option	Exact 1	≥ 0	≥ 0			
Missing Item Option	Error	Ignore	Ignore			

Notes: (1) AGC, LGC, AGS

(2) See Flowchart of GMSMED for storing details.

MED Code: P16

Purpose: Generate an ephemeris for one vehicle using a vector from the other vehicle

Example: P16,CSM,LEM;

Item Desc.	1	2	3	4	5	6
Item Name	Old vehicle	New vehicle	GMT	Maneuver number		
Input Format	EBCDIC	EBCDIC	h:m:s(.th)	FXP		
Input Units	EBCDIC	EBCDIC	hours	FXP		
Checking Option	Exact	Exact	≥ 0	≥ 0		
Missing Item Option	Error	Error	Insert zero	Insert zero		

Notes: (1) Vehicle must be CSM, LEM

(2) GMT and maneuver are mutually exclusive (i.e., must be one but not both).

MED Code: P51

Purpose: Offsets and elevation angle for two-impulse solution

Example: P51,15,-4,26.6,130;

Item Desc.	1	2	3	4	5	6
Item Name	Delta Height	Phase Angle	Elevation Angle of Target	Travel Angle for Terminal Phase		
Input Format	FLP	FLP	FLP	FLP		
Input Units	NM	deg.	deg.	deg.		
Checking Option	None	None	None	None		
Missing Item Option	Ignore	Ignore	Ignore	Ignore		

Notes:

MED Code: P52

Purpose: Two-impulse corrective combination nominals

Example: P52,28:00:00,8,-1.32;

Item Desc.	1	2	3	4	5	6
Item Name	Nom. Time of NSR maneuver	Nom. Height Difference at NSR	Nom. Phase Angle at NSR			
Input Format	H:M:S(.TH)	FLP	FLP			
Input Units	hours	NM	deg.			
Checking Option	≥ 0	None	None			
Missing Item Option	Error	Error	Error			

MED Code: P80

Purpose: Initialize number of vehicles, first launch vehicle, mission date

Example: P80,1,CSM,7,16,1969;

Item Desc.	1	2	3	4	5	6
Item Name	Number of vehicles	First Launch Vehicle	Month	Day	Year	Delta Day
Input Format	FXP	EBCDIC	EBCDIC	FXP	FXP	FXP
Input Units	NA	EBCDIC	EBCDIC	NA	NA	NA
Checking Option	N = 1	Exact 1	Exact 2	$1 \leq D \leq 31$	$50 \leq Y \leq 1980$	≥ 0
Missing Item Option	Note 4	Note 5	Note 4	Note 4	Note 4	Note 4

Notes: (1) Vehicle must be in MHGVNM.

(2) Date is checked by internal calendar on final logic. Then reference day is calculated (Jan 1 = day 0) along with days in month, etc. These values are stored in GZGENCSN.

(3) On entry, link to EMLAMPNP to rotate P and N matrices and sun/moon tables.

(4) If in Nophase, items 1-5 required, 6 may be input (zero inserted if missing). If in any phase, items 1-5 must be missing, and item 6 must be input.

MED Code: U00

Purpose: Space digitals initialization

Example: U00,CSM;

Item Desc.	1	2	3	4	5	6
Item Name	VEH ID	CENTRAL BODY				
Input Format	VEH	A				
Input Units	EBCDIC	EBCDIC				
Checking Option	Exact (1)	Exact(2)				
Missing Item Option	Error	Assume "E"				

Notes: (1) CSM or LEM

(2) E for Earth (=1)

M for moon (=3)

(3) EBCDIC name and numeric code

MED Code: U01

Purpose: Space digitals

Example: U01,1,GET,100:00;00;

Item Desc.	1	2	3	4	5	6
Item Name	MANUAL COLUMN	OPTION IND	PARAM- ETER	INCLINA- TION	ASCENDING NODE	
Input Format	N	AAA	HHH:MM:SS OR NN	NNN.NNN	NNN.NNN	
Input Units	FXP	EBCDIC	hours or FXP	deg	deg	
Checking Option	MINMAX(1)	Exact(2)	None	MINMAX(4)	None	
Missing Item Option	Error	Error	Error	None(3)	None(3)	

Notes: (1) $1 \leq N \leq 3$
(2) GET or MNV
(3) Mandatory when manual column = 2, otherwise illegal
(4) 0° to 180°

MED Code: U02

Purpose: Initiate checkout monitor

Example: U02,CSM,GET,100:00:00,90:00:00,ECT,FT;

Item Desc.	1	2	3	4	5	6
Item Name	Veh Id	Option Ind	Parameter	Threshold Time	Reference	Feet
Input Format	Veh	AAA	HHH:MM:SS OR NN	HHH:MM:SS	AAA	AA
Input Units	EBCDIC	EBCDIC	Hours, FXP or FLP	Hours	EBCDIC	EBCDIC
Checking Option	Exact(2)	Exact(1)	Special(3)	None	Exact(5)	Exact(7)
Missing Item Option	Error	Error	Error	Special(4)	Insert(6)	Insert zero

Notes:

<u>IND</u>	<u>IND CODE</u>	<u>PARAMETER</u>
	1	TIME
	2	TIME
(1) MVI	3	FXP MNV. NO.
MVE	4	FXP MNV. NO.
RAD	5	FLP RADIAL CUTOFF
ALT	6	FLP ALTITUDE CUTOFF
FPA	7	FLP FLIGHTPATH ANGLE CUTOFF

(2) CSM or LEM

(3) Parameter must be consistent with option ind.

(4) Optional for GET, GMT, illegal for MVI, MVE, mandatory for RAD, ALT, FPA

(5) ECI=0, ECT=1, MCI=2, MCT=3

(6) Assume ECI(=0)

(7) FT

(8) Reference indicator set negative if FT input

MED Code: U07

Purpose: Moonrise/Moonset Display

Example: U07,GET,100:00:00;

Item Desc.	1	2	3	4	5	6
Item Name	IND	PARAM				
Input Format	AAA	HHH:MM:SS or NN				
Input Units	Note 2	hours or FXP				
Checking Option	Exact(1)	none or MIN- MAX(4)				
Missing Item Option	Error	Error(3)				

Notes: (1) GET if time to be input, REV if REV to be input

(2) GET or REV

(3) Insert zero

(4) Current REV $\leq$ REV $\leq$ Last REV in Cape Table

MED Code: U08

Purpose: Sunrise/Sunset Display

Example: U08,GET,100:00:00;

Item Desc.	1	2	3	4	5	6
Item Name	IND	PARAM				
Input Format	AAA	HHH:MM:SS or NN				
Input Units	Note 2	hours or FXP				
Checking Option	Exact(1)	none or MIN- MAX(4)				
Missing Item Option	Error	Error(3)				

Notes: (1) GET if time to be input, REV if REV to be input

(2) GET or REV

(3) Insert zero

(4) Current REV $\leq$ REV $\leq$ Last REV in Cape Table

MED Code: U12

Purpose: Predict apogee/perigee (FDO Orbit Digitals)

Example: U12,CSM,GET,100:00:00;

Item Desc.	1	2	3a	3b	3c	4
Item Name	VEH ID	IND	REV NO	TIME	MNV NO	CENTRAL BODY
Input Format	VEH	AAA	N	HHH:MM:SS	NN	A
Input Units	EBCDIC	EBCDIC	FXP	GMT	FXP	EBCDIC
Checking Option	Exact(3)	Exact(1)	MINMAX(2)	None	None	Exact(4)
Missing Item Option	Error	Error	Error	Error	Error	Assume "E" (5)

Notes: (1) IND = REV, parameter 3a entered
= GET, parameter 3b entered
= MNV, parameter 3c entered
(2) Current REV $\leq$ REV $\leq$ Last REV in Cape Table
(3) CSM or LEM
(4) E for Earth (=0)
(5) This parameter is valid only when IND=REV

MED Code: U13

Purpose: Longitude crossing times (FDO Orbit Digitals)

Example: U13,CSM,1,90;

Item Desc.	1	2	3	4	5	6
Item Name	VEH ID	REV NO	Longitude	CENTRAL BODY		
Input Format	VEH	NN	+DDD.XXXX	A		
Input Units	EBCDIC	FXP	LONG	EBCDIC		
Checking Option	Exact(2)	MINMAX(1)	None	Exact(3)		
Missing Item Option	Error	Error	Error	Assume "E"		

Notes: (1) Current REV $\leq$ REV $\leq$ Last REV in Cape Table

(2) CSM or LEM

(3) E for Earth (=0)

MED Code: U14

Purpose: Compute longitude at given time (FDO Orbit Digitals)

Example: U14,CSM,100:00:00;

Item Desc.	1	2	3	4	5	6
Item Name	VEH ID	TIME				
Input Format	VEH	HHH:MM:SS				
Input Units	EBCDIC	GET				
Checking Option	Exact(1)	None				
Missing Item Option	Error	Error				

Notes: (1) CSM or LEM

MED Code: U20

Purpose: Generate detailed maneuver table display

Example: U20,CSM,1;

Item Desc.	1	2	3	4	5	6
Item Name	MPT ID	Maneuver Number	MSK Number	REFSMMAT	Heads up/ Heads down	
Input Format	AAA	NN	NN	AAA	A	
Input Units	EBCDIC	FXP	FXP	EBCDIC	D,U	
Checking Option	Exact(1)	MINMAX(2)	Exact(3)	Exact(4)	Note (5)	
Missing Item Option	Error	Error	assume 54	assume CUR	Note (6)	

Notes: (1) CSM or LEM

(2) $1 \leq NN \leq 15$

(3) 54 or 69

(4) CUR = 1, PCR = 2, TLM = 3, OST = 4, MED = 5, DMT = 6, DOD = 7, LCV = 8, DES = 9, LLA = 10, LLD = 11

(5) If item 4 $\neq$ DES, item 5 must not be input

(6) Assume U if item 4 = DES and item 5 is missing

13 MOCR Displays

13.1 FDO Launch Analog No. 1 (MSK 0040)

The purpose of the FDO Launch Analog No. 1 is to serve as the primary display for trajectory evaluation during launch. The inertial flight-path angle (γ) in degrees versus the inertial velocity (V) in feet per second is plotted on a half-second cycle.

13.2 FDO Launch Analog No. 2 (MSK 0041)

The purpose of the FDO Launch Analog No. 2 is to display conditions of $(\gamma, V)_{EI}$. The flight-path angle versus inertial velocity at entry interface (γ vs $V)_{EI}$ is plotted. The plot is initialized when apogee altitude is equal to or above entry interface. It is then updated and terminated at reaching orbit.

13.3 FDO Launch Digital No. 1 (MSK 0043)

TBD.

13.4 FDO Orbit Digitals (MSK 0045 and 0046)

13.4.1 Function

The function of the Flight Dynamics Officer (FDO) Orbit Digitals Display is to compute present position information and predicted data concerning apogee/perogee and longitude crossings as manually requested.

13.4.2 Display Parameters

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
Vehicle	Vehicle name		
Rev	Current revolution number associated with subject vehicle and central body		(1)
Vector ID	Last vector used for updating the ephemeris		(2)
GMT ID	GMT Time of anchor vector	Hr:min:sec	(2)
GET ID	GET Time of anchor vector	Hr:min:sec	(2)
GET	Current ground elapsed time	Hr:min:sec	(1)
Weight	Total current weight of subject vehicle	lbs.	(1)
K-Factor	Atmospheric density multiplier considered in generating the ephemeris		(1), (2)
λ_{PP}	Present position, longitude (Seleno- graphic for moon)	deg.	(1), (2)
ϕ_{PP}	Present position, latitude (geodetic for earth, Selenographic for moon)	deg.	(1), (2)
GET _{CC} *	GET of arrival at next cape crossing	Hr:min:sec	(1), (2)

(1) Output cycle (12 sec.)

(2) Trajectory Update, MSK Request

* For lunar orbit this will be the time of the vehicle crossing of the 180deg selenographic longitude.

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
N_{PP}	True anomaly	deg.	(1), (2)
λ_{AN}	Longitude of the ascending node (earth fixed or moon fixed)	deg.	Node crossing, (2)
h	Current oblate height (Spherical for moon)	N.M.	(1), (2)
V_i	Current inertial velocity	f.p.s.	(1), (2)
γ	Current inertial flightpath angle	deg.	(1), (2)
a	Semimajor axis of orbital ellipse	N.M.	(1), (2)
e	Eccentricity of orbital ellipse		(1), (2)
i	Orbital inclination to central body equator	deg.	(1), (2)
GET_{EI}	GET of arrival at unsafe altitude	Hr:min:sec	(2)
ϕ_{EI}	Geodetic latitude of entry interface	deg.	(2)
λ_{EI}	Longitude of entry interface	deg.	(2)
PET	Phase elapsed time (GET of an event minus GETR)	Hr:min:sec	(4)

(1) Output cycle (12 sec.)

(2) Trajectory Update, MSK Request

(4) These times will be updated when a new GETR and a new event is specified.

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
GETR	GET of reference (elapsed time of an event)	Hr:min:sec	(4)
GET _λ	Time S/C will pass over λ below	Hr:min:sec	Computed or manually entered
REV _λ	Revolution associated with GET _λ		Manually entered or computed
λ	The longitude associated with above GET	deg.	Manually entered or computed
h _{a**}	Spherical height of next apogee at GET _a	N.M.	Crossing of Apogee, (2), maneuver executed
φ _{a**}	Latitude of next apogee at GET _a . Geocentric in earth reference, seleno- graphic around moon	deg.	Crossing of Apogee, (2), maneuver executed
λ _{a**}	Longitude of next apogee at GET _a	deg.	Crossing of Apogee, (2), maneuver executed
GET _{a**}	Time of arrival at next apogee	Hr:min:sec	Crossing of Apogee, (2), maneuver executed
h _{p**}	Spherical height of next perigee at GET _p	N.M.	Crossing of Perigee, (2), maneuver executed
φ _{p**}	Latitude at next perigee at GET _p . Geocentric relative to the earth , selenographic around moon	deg.	Crossing of Perigee, (2), maneuver executed
λ _{p**}	Longitude of next perigee at GET _p	deg.	Crossing of Perigee, (2), maneuver executed
GET _{p**}	Time of arrival at next perigee	Hr:min:sec	Crossing of Ppogee, (2), maneuver executed
GET _{BV}	Time tag of vector from which apogee/perigee values were computed	Hr:min:sec	Computed or manually entered
REV _{BV}	Revolution of GET _{BV}		Computed or manually entered

(2) Trajectory Update, MSK Request

(4) These times will be updated when a new GETR and a new event is specified.

** Also displayed manually for future time periods as requested.

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
T_0	Orbital Period	Hr:min:sec	(1)
REF1, REF2, REF3	Indicator of which central body is being referenced for present position data and for manual Apogee/Perigee and longitude		
NV_1	Number of vectors used for interpolation for present position values		
NV_2	Number of vectors used in interpolation for base vector for predicted apogee/perigee data		
Trajectory Update Number	Update number associated with subject vehicle ephemeris		(2)

13.5 Mission Plan Table (MSK 0047)

13.5.1 Function

13.6 FDO Detailed Maneuver Table (MSK 0054 and 0069)

13.6.1 Function

The Detailed Maneuver Table Display module provides detailed maneuver information for any specified maneuver in either the CSM or LM Mission Plan Table. Two maneuvers may be evaluated by simultaneous display of a DMT for each maneuver. The information displayed is obtained as follows:

1. Maneuver execution parameters, including times, DV quantities, vehicle attitudes and weights, are picked up from the appropriate MPT block. IMU attitudes and VG are computed based on a specified reference matrix. FDAI angles for the LM are computed from IMU attitudes (FDAI and IMU angles will be the same for CSM).
2. Resultant trajectory parameters are also picked up from the appropriate MPT block. When two ephemerides exist at cutoff, rendezvous-related quantities are computed by the DMT.
3. Targeting parameters are displayed when applicable. The following are the available guidance types for the indicated on-board computers:
 - (a) IU - Translunar injection IGM
 - (b) CMC - Lambert or external DV
 - (c) LGC - Lambert or external DV
 - (d) AGS - External DV
4. For unfrozen maneuvers where the LM primary external DV guidance mode is specified, LM AGS external DV target parameters are computed and displayed as backup guidance parameters. The parameters remain unchanged after the maneuver is frozen.

13.6.2 Display

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
C STA ID	standard identification for CSM vector used in generation of maneuver	
GMTV	Greenwich mean time of CSM vector	hr:min:sec
GETV	Ground elapsed time of CSM vector	hr:min:sec
CODE	code assigned to the maneuver in the MPT	
L STA ID	standard identification for LM vector used in generation of maneuver	
GMTV	Greenwich mean time of LM vector	hr:min:sec
GETV	Ground elapsed time of LM vector	hr:min:sec
REF	prints out either EARTH or LUNAR depending on the reference body	
X STA ID	standard identification for the CSM or LM vector that was used to generate the frozen profile of the maneuver displayed in the DMT	
GMTV	Greenwich mean time of frozen maneuver vector	hr:min:sec
GETV	Ground elapsed time of frozen maneuver vector	hr:min:sec
GETR	Reference ground elapsed time for phase elapsed time	hr:min:sec
WT	Weight of the total maneuvering vehicle prior to the maneuver	lb
WC	CSM total weight prior to the maneuver	lb
WL	LM total weight prior to the maneuver	lb
WF	Weight prior to the maneuver of the engine performing the maneuver	lb
GETI	Ground elapsed time of ignition	hr:min:sec
PETI	Phase elapsed time of ignition	hr:min:sec
DTB	Duration of main engine burn	min:sec
DTU	Duration of ullage	sec
DTTO	Effective duration of main engine tailoff	sec
DVTO	Velocity increment of main engine tailoff	fps
REFSMMAT	Identification of REFSMMAT used to compute gimbal angles	
DP	Main engine pitch gimbal trim angle from the electronic null position for the SPS or the positive handover position for the DPS	deg
DY	Main engine yaw gimbal trim angle from the electronic null position for the SPS or the positive handover position for the DPS	deg

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
DVM	Total velocity increment of maneuver	fps
DVREM	Total velocity remaining after maneuver	fps
DVC	Total velocity increment, less tailoff, to the applied along the vehicle X-axis	fps
VGX, VGY, VGZ	Prethrust VG output from the guidance equations displayed in IMU coordinates. Computed based on a free-flight vector at GETI	fps
OY, IP, MR	Outer, inner and middle gimbal angles at ignition (0° - 360°). These labels will be dynamic and displayed as follows: for the CSM, roll pitch, yaw. For the LM: Yaw, pitch, roll. For the S-IVB maneuver these will be CSM gimbal angles	deg
YP, PB, RB	FDAI angles. These are the same as the gimbal angles yaw, pitch, roll for CSM maneuvers, but correspond to pilot yaw, pitch, roll for LM maneuvers	deg
YH, PH, RH	Local vertical-local horizontal yaw, pitch and roll at ignition	deg
VF, VS, VD	Components of VGX, VGY, VGZ transformed to local vertical coordinates defined at GETI	fps
ΔH	Differential altitude at cutoff between LM (S-IVB) and CSM, positive when maneuver vehicle is below	NM
θ	Phase angle at cutoff between LM (S-IVB) and CSM measured from chaser to target in the direction of motion (0°-360°)	deg
$\dot{\theta}$	Rate of change of phase angle at cutoff, positive when phase angle is decreasing	deg per orbit
δ	Crossrange distance from CSM orbit to LM at insertion following ascent. Will be zero for any maneuver other than the ascent maneuver	NM
H _{BI}	Altitude of maneuvering vehicle above spherical earth or moon (defined by REF) at ignition time	NM
ϕ_{BI}	Geodetic latitude of maneuver vehicle at ignition while in earth orbit. Otherwise it is geocentric latitude	deg
λ_{BI}	Longitude of maneuver vehicle at ignition	deg
ν_{BI}	True anomaly of maneuver vehicle at ignition	deg

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
UNTIL	Lighting conditions at GETI	hr:min until day/night
h_a	Altitude of next apoapsis after cutoff above spherical reference	NM
h_p	Altitude of next periapsis after cutoff above spherical reference	NM
LAN	Longitude of next ascending node after cutoff	deg:min
E	Orbital eccentricity at cutoff	
I	Trajectory inclination to (earth or moon defined by REF) resulting from maneuver	deg
ω_p	Argument of perigee or pericyynthion (defined by REF) resulting from maneuver	deg
VP	Velocity at next pericyynthion after maneuver for maneuver performed while in lunar orbit	fps
θ_p	Central angle between radius vector passing through the next pericyynthion after cutoff and the lunar landing site at the time of next pericyynthion and negative when the landing site is beyond pericyynthion; driven only for maneuvers while in lunar orbit	deg
δ_p	Wedge angle between orbit at cutoff and a parallel plane passing through the lunar landing site; driven only for maneuver while in lunar orbit	deg
ϕ_{LLS}	Latitude of lunar landing site as a result of the powered landing time	deg:min
λ_{LLS}	Longitude of lunar landing site as a result of the powered landing time	deg:min
RLLS	Radius of lunar landing site	NM

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
<u>TARGETS</u>	These quantities are driven according to the type of maneuver displayed. They are as follows:	
PGNS EXT DV	Primary guidance and navigation system ID for vehicle doing the external DV maneuver	
PGNS GETI	Ground elapsed time of ignition	hr:min:sec
PGNS VX, VY, VZ	Components of external DV vector required for targeting the onboard PGNS external DV guidance for maneuver execution	fps
AGS EXT DV	Abort Guidance System ID for vehicle doing the external DV maneuver	
AGS GETI	Ground elapsed time of ignition	hr:min:sec
AGS VX, VY, VZ	Components of external DV vector required for targeting the onboard AGS external DV guidance for maneuver execution	fps

13.7 Space Digitals (MSK 0082)

13.7.1 Function

The function of the Space Digitals Display is to compute and display parameters necessary to evaluate and monitor trajectories involving Earth-moon relationships. At the top of the display present position data are shown. In column 1 orbital elements at an input time are displayed. Column 2 has the time of lunar sphere of influence entry, numbers for the closest approach to the Moon and at the node, the intersection with the lunar orbit plane (TBD). The third column shows the time of lunar sphere of influence exit, the closest approach to Earth after that exit and, if applicable, the state at Earth entry interface.

13.7.2 Display

The update associated with this display are:

- (1) Time cycle (12 sec)
- (2) Trajectory update
- (3) Manual entries (3)
- (4) MSK request
- (5) Reinitialization

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
Vector-ID	Identification of the last vector used to update the ephemeris	None	(1), (2), (4), (5)
Weight	Total vehicle weight	lbs	(1), (2), (4), (5)
GMTV	Greenwich time-tag of the vector	HHH:MM:SS.SS	(1), (2), (4), (5)
GETV	Ground elapsed time-tag of the vector	HHH:MM:SS.SS	(1), (2), (4), (5)
GET Axis	Ground elapsed time used to define the earth-moon line in the initialization of the earth-moon transit display	HHH:MM:SS.SS	(1), (2), (4), (5)
GETR	Ground elapsed time reference (elapsed time of an event)	HHH:MM:SS.SS	(1), (2), (4), (5)
GET	Current ground elapsed time for which V, GAM, PHI, LAM, H, PSI and ADA were computed	HHH:MM:SS.SS	(1)
REF	Inertial reference body used to compute V, GAM, PHI, LAM, H, PSI and ADA	HHH:MM:SS.SS	(1)
V	Current velocity (earth centered inertial or lunar centered inertial)	ft/sec	(1), (2), (4), (5)

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
GAM	Current inertial flightpath angle (earth or lunar reference)	deg	(1), (2), (4), (5)
H	Current altitude above an oblate earth or spherical above the moon assuming landing site radius	nm	(1), (2), (4), (5)
PHI	Current geodetic latitude or current selenographic latitude respect to a spherical moon whose radius is the landing site	deg	(1), (2), (4), (5)
LAM	Current earth longitude or current selenographic longitude	deg	(1), (2), (4), (5)
PSI	Current heading angle with respect to the earth or moon, measured clockwise from north	deg	(1), (2), (4), (5)
ADA	Current true anomaly	deg	(1), (2), (4), (5)
VEH ID	Vehicle for which the space digitals are computed	XXX	(2), (3), (4), (5)
GET Vector 1	Ground elapsed time of the vector used to compute the quantities below	HHH:MM:SS.SS	(3)
REF	The inertial reference body used to compute the quantities (i.e., if GET Vector 1 is inside the lunar sphere of influence, REF is MOON)	None	(3)
WT	Total weight at time of GET Vector 1	lbs	(3)
GETA	Ground elapsed time of next apogee (reference from GET Vector 1)	HHH:MM:SS.SS	(3)
HA	Height of apogee above a spherical earth or height of apolune above a spherical moon whose radius is the landing site (referenced from GET Vector 1)	nm	(3)
HP	Height of perigee above a spherical earth or height of perilune above a spherical moon whose radius is the landing site (referenced from GET Vector 1)	nm	(3)

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
IEMP	Inclination of the trajectory with respect to the earth-moon plane (referenced from GET Vector 1)	deg	(3)
HS	Altitude above a spherical earth or moon (referenced from GET Vector 1)	nm	(3)
PHI	Geocentric latitude or selenographic latitude with respect to a spherical moon of landing site radius (referenced from GET Vector 1)	deg	(3)
OMG	Right ascension of the ascending node (inertial)	deg	(3)
PRA	Inertial right ascension of perigee	deg:min	(3)
K	Density multiplier used in the computation of the above quantities	None	(3)
GET Vector 2	Ground elapsed time of the vector supplied by the ephemeris and from which the quantities below will be computed	HHH:MM:SS.SS	(3)
GETSI Sphere Entrance	The first ground elapsed time after GET Vector 2 that the vehicle will pass through the lunar sphere of influence with a negative lunar altitude rate (rate with respect to the moon is decreasing)	HHH:MM:SS.SS	(3)
GETCA	Ground elapsed time of closest approach to the moon	HHH:MM:SS.SS	(3)
VCA	Velocity with respect to a moon-centered inertial coordinate system at the point of closest approach to the moon	ft/sec	(3)
HCA	Altitude at the point of closest approach to a spherical moon of landing site radius	nm	(3)
PSICA	Heading angle at the point of closest approach to the moon measured clockwise from north	deg	(3)
PHICA	Selenographic latitude with respect to a spherical moon at the point of closest approach to the moon	deg	(3)

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
LCA	Selenographic longitude with respect to the moon	deg	(3)
GETMN	Ground elapsed time of arrival at the node defined by the planes of the approach hyperbola and the desired lunar parking orbit	HHH:MM:SS.SS	(3)
PMN	Selenographic latitude relative to a spherical moon at GETMN	deg	(3)
LMN	Selenographic longitude relative to a spherical moon at GETMN	deg	(3)
HMN	Selenographic height relative to a spherical moon of landing site radius	nm	(3)
DMN	Wedge angle between the planes of the approach hyperbola and the desired lunar parking orbit (Note: A positive sign indicates that the inclination of the desired lunar parking orbit is less than the inclination of the approach hyperbola)	deg	(3)
GET Vector 3	Ground elapsed time of the vector supplied by the ephemeris and from which the quantities below will be computed	HHH:MM:SS.SS	(3), (2)
GETSE Sphere Exit	The first ground elapsed time after GET Vector 3 that the vehicle will pass through the lunar sphere of influence with a positive lunar altitude rate (Note: Altitude with respect to the moon is increasing)	HHH:MM:SS.SS	(3), (2)
GETEI	Ground elapsed time of entry interface	HHH:MM:SS.SS	(3), (2)
VEI	Earth centered velocity at entry interface	ft/sec	(3), (2)
GEI	Inertial flightpath angle at entry interface	deg	(3), (2)
PEI	Geodetic latitude at entry interface	deg	(3), (2)
LEI	Earth longitude at entry interface	deg	(3), (2)

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>	<u>UPDATES</u>
PSIEI	Heading angle with respect to earth at entry interface	deg	(3), (2)
GETVP	Ground elapsed time of arrival at vacuum perigee	HHH:MM:SS.SS	(3)
VVP	Earth centered velocity at vacuum perigee	ft/sec	(3)
HVP	Altitude above an oblate earth at vacuum perigee	nm	(3)
PVP	Geodetic latitude at vacuum perigee	deg	(3)
LVP	Earth longitude at vacuum perigee	deg	(3)
PSIVP	Heading angle with respect to earth at vacuum perigee	deg	(3)
IE	Inclination angle with respect to the earth at vacuum perigee	deg	(3)
LN	Geographic longitude of the earth return ascending node	deg	(3)

13.8 CSM Optics Support Table (MSK 0229)

TBD

13.9 Ascent Rendezvous Monitor (MSK 0232)

The Ascent Rendezvous Monitor (ARM) display calculates a tweak burn to be performed shortly after insertion from lunar ascent to correct the trajectory and get back to the nominal coelliptic rendezvous profile. It also shows additional parameters for contingencies, like the time of ignition and Delta V of an apolune kick burn, to correct an unsafe orbit.

13.9.1 Inputs

See input page in the MFD.

13.9.2 Display

The display automatically updates every two seconds after it has been enabled with the ENA button. This can be done before lunar liftoff, although no reasonable calculations can be done until shortly before insertion. Before that error messages will be displayed.

According to the mission rules the tweak burn is required if its DV exceeds 30 ft/s. For this recommended action is display which either says GO or NO GO. The apolune kick calculation is done if the current perilune is lower than the minimum safe altitude from the input page.

List of error codes:

1. Failure in conic state vector extrapolation routine
2. CDH indicator must be an odd number
3. Too many iterations

13.10 Short Ascent Rendezvous Monitor (MSK 0233)

The Short Ascent Rendezvous Monitor (Short ARM) display is, like MSK 232, computing a tweak burn to be performed shortly after lunar ascent, but it is used for the direct/short rendezvous profile employed on Apollo 14 and later. On these missions the ARM display is used to calculate a bailout burn to get back to a coelliptic profile.

13.10.1 Inputs

See input page in the MFD.

13.10.2 Display

The display automatically updates every two seconds after it has been enabled with the ENA button. This can be done before lunar liftoff, although no reasonable calculations can be done until shortly before insertion. Before that a numeric error code will be displayed.

According to the mission rules the tweak burn should not be performed if the DV exceeds 60 feet per second or if the perilune altitude after the burn would be below 5 nautical miles. The recommendation message on the display shows either BAIL or TWEAK for this. No specific recommendation is given if the tweak burn Delta V is so small that it should not even be performed.

List of error codes:

1. Failure in Two-Impulse processor

13.11 LM Optics Support Table (MSK 0239)

The LM Optics Support Table is used to evaluate orbital IMU alignments and provide ground support for alignment checks, docked alignments and backup AGS alignments. Several modes are available. In all modes a realignment attitude (IMU and FDAI) is calculated. This attitude is referenced to matrix 1 and gives the body alignment (i.e. gimbal angles all zero) with matrix 2. The star codes are in decimal, unlike the AGC stars which are in octal. The RTCC star catalog is given later, which can be used to convert between the two systems.

13.11.1 DOK

In the docking alignment mode either a REFSMMAT, CSM or LM attitude are calculated. There are three options: option 1 computes the LM REFSMMAT, option 2 computes the LM attitude and option 3 computes the CSM attitude. A CSM REFSMMAT needs to be specified as well as the two other data sets (attitudes or LM REFSMMAT) that are not being calculated.

13.11.2 AGS

The AGS backup alignment requires the AOT to be placed on a star in any detent and the cursor being placed on a second star in the same detent. Attitudes can be entered, but should be set to zero due to the AGS body alignment mode. For star 1 the star number and detent have to be specified. For the second star the number, reticle angle and the cursor used have to be entered. The result is an optics (OST) REFSMMAT being calculated as matrix 1. If then the desired REFSMMAT is entered as matrix 2 the realignment attitude is calculated relative to matrix 1, that will give the desired alignment relative to matrix 2.

13.11.3 CHK

The alignment check mode can only be used with the MPT being initialized. An attitude can be input, for example a burn attitude. The display then calculates two check stars for the AOT and two stars for the COAS, if available.

13.11.4 MAT

The matrix mode is only used to view any REFSMMAT in the LM locker.

13.11.5 LUN

Not implemented yet.

13.11.6 FLT

Not implemented yet.

13.11.7 Buttons

MOD: Choose the mode of the display. This should be done first. Valid options right now are DOK, AGS, MAT (no special processing, only to show REFSMMATs) and CHK (only works with initialized MPT). LUN and FLT modes not implemented yet.

AT1: Enter the first set of attitudes. For the DOK mode this is the LM attitude, for the AGS mode it is the initial attitude (should always be set to zero though) and for the CHK mode it is the attitude at which the star checks are calculated, for example a burn attitude.

AT2: Enter the second set of attitudes. For the DOK mode this is the CSM attitude, for the AGS mode the attitude at which the second star data was taken (should always be set to zero though).

CRF: Choose the type of CSM REFSMMAT used in the DOK mode.

MA1: Choose the LM REFSMMAT that is displayed as matrix 1 of the display. In the DOK mode this is the REFSMMAT used in the calculation of the CSM or LM attitude. With the realignment option this is the current REFSMMAT after an AGS body alignment was done using the AGS alignment technique.

MA2: Choose the LM REFSMMAT that is displayed as matrix 2 of the display. For the realignment option this REFSMMAT is the one that is desired after the second AGS body alignment, at the calculated attitude referened to matrix 1.

CLC: Calculate the display parameters according to the chosen mode.

OP1: Enter optics data for the first star. In the AGS mode this is the star at the center of the AOT and the star ID and detent have to be given.

OP2: Enter optics data for the second star. In the AGS mode this is the star on the cursor and the star ID, detent position (which axis is used as the cursor) and the cursor angle have to be given.

REA: Calculate the realignment attitudes (IMU and FDAI). The realignment attitudes will give angles in the coordinate system, described by MAT 1, to align the spacecraft body axes to the coordinate system, described my MAT 2.

BCK: Go back to previous display.

13.11.8 Display Parameters

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
VEH	Vehicle type. Allows multiple LM-type vehicle capability.	
MODE	Indicates type of computations for which the display is being used	
DKAN	The docking angle obtained from the interface of the LM and CSM docking hatches, indicating the relative rotation of one vehicle form the other about the spacecrafts' X axes	DEG
LM YPR	LM attitude used in the DOK mode	DEG
CSM RPY	CSM attitude used in the DOK mode	DEG
MATRIX USED	The CSM REFSMMAT used in docked alignment computations if display is in docked mode;the LM REFSMMAT used in optics computations for an alignment check if display is in check mode; or the REFSMMAT indicating the desired alignment for the AGS if the display is in the AGS mode	
GETHORIZ	Ground elapsed time of alignment check by star sightings and/or horizon sighting	HH:MM:SS
OGA	The outer gimbal angle required to put the spacecraft plus Z axis line-of-sight tangent to the horizon at GETHORIZ	DEG
S1 and S2	AOT stars used or to be used in optical sightings	
RA	Right ascension of star	DEG
DEC	Declination of star	DEG
YPR	Yaw, pitch and roll LM gimbal angles or AGS angles used in alignment checks. The attitudes will be those related to star-sighting data in CHK, LUN and AGS modes	DEG
L	AOT reticle line to be superimposed with star S1 or S2	Plus or minus X- or Y-axis
A1	The AOT reticle angle counter position to superimpose target star and Line L	DEG
A2	The AOT reticle angle counter position to superimpose target star and spirale	DEG
D	AOT detent position, 1 through 6	

<u>QUANTITY</u>	<u>DEFINITION</u>	<u>DIMENSIONS</u>
S1 and S2	COAS stars used or to be used in optical sightings	
EL	The COAS elevation angle ranging between plus and minus 40 degrees to place star on horizontal X-line. A positive angle displaces the COAS LOS counter-clockwise about the LM +Y axis	DEG
SXP	The Star-X-Position coordinate ranging between plus and minus 5 degrees. The right side of the reticle is plus giving a star on the +Y side of the LM X-Z plane a positive SXP	DEG
REALIGN ATTITUDES	The attitudes in the coordinate system described by MAT 1 to align the spacecraft body axis to the coordinate system described by MAT 2. Attitudes will be given as gimbal angles and FDAI angles	DEG
GIMB ANG	Realign attitudes specified in gimbal angles	DEG
FDAI ANG	Realign attitudes specified in FDAI angles	DEG
STORED AT	Telemetry and MED attitudes stored by the LOST program	DEG
MATRIX1 and 2	Any two matrices stored in the LM REFSMMAT locker, and their code identifications, which are to be displayed in positions 1 and 2 on the LOST display	
MAT GET	An associated GET of any matrix display, such as time computed, time made current or time saved (the most meaningful time will be used in each case)	HH:MM:SS

(a) RTCC star catalogue for J2000.0 epoch

Star no.	IAU designation	Name	Relative brightness and constellation ⁴	Common name	Right ascension, hours:minutes:seconds	Declination, degrees:arcminutes:arcseconds	Magnitude	Star no.	IAU designation	Name	Relative brightness and constellation ⁴	Common name	Right ascension, hours:minutes:seconds	Declination, degrees:arcminutes:arcseconds	Magnitude	Star no.	IAU designation	Name	Relative brightness and constellation ⁴	Common name	Right ascension, hours:minutes:seconds	Declination, degrees:arcminutes:arcseconds	Magnitude	
1	a And	a Andromedae	Alpheratz	06:06:49.9	+28:55:29	2.1	1	Alpheratz								71	n Osa	n Canis Majoris	Aludra	7:02:54.5	-29:14:37	2.4	ALUDRA	ETA CANIS MAJORIS
2	8 Cet	8 Ceti	Diphda	04:02:05.0	-18:09:04	2.2	2	Diphda								72	c Pup	c Puppis	Shaula	8:02:13.8	-39:55:04	2.3	SHAULA	ZETA PUPII
3	y Cas	y Cassiopeiae	Waz	01:24:53.0	+46:03:17	2.2	3	Waz								73	a Vel	a Velorum	Subail	9:06:53.5	-43:18:39	2.2	SUBAIL	LAMBDA VELORUM
4	a Eri	a Eridani	Achernar	1:36:35.9	-57:23:20	0.6	4	Achernar								74	i Car	i Carinae	Misak	9:16:17.3	-59:08:57	2.2	MISAK	TOTA CARINAE
5	a UMi	a Ursae Minoris	Polaris	2:03:18.9	+89:07:34	2.1	5	Polaris								75	8 UMa	8 Ursae Majoris	Misak	11:00:02.5	+45:32:37	2.4	MISAK	BETA URSAE MAJORIS
6	8 Eri	8 Eridani	Menkar	2:57:07.4	-40:25:27	3.4	6	Menkar								76	c UMi	c Ursae Minoris	Kochab	13:52:43.5	+45:04:53	2.4	KOCHAB	BETA URSAE MINORIS
7	a Cet	a Ceti	Menkar	3:00:42.5	+3:58:23	2.8	7	Menkar								77	8 UMa	8 Ursae Majoris	Eltanin	14:50:46.5	+74:16:41	2.2	ELTANIN	GAMMA DRACONIS
8	a Per	a Persei	Mirfak	3:22:10.3	+49:45:21	1.9	8	Mirfak								78	y Cyg	y Cygni	Sadr	17:55:54.5	+51:29:30	2.4	SADR	GAMMA CYGNI
9	a Tau	a Tauri	Rigel	4:34:11.8	+21:27:01	1.1	9	Rigel								79	a Gru	a Gruis	Al Na'ir	22:06:20.9	-47:06:26	2.2	AL NA'IR	ALPHA GRUI
10	8 Ori	8 Orionis	Rigel	5:13:09.7	-8:14:06	0.3	10	Rigel								80	8 Gru	8 Gruis	Schedar	22:06:52.9	-47:02:32	2.2	SCHEDAR	BETA GRUI
11	a Aur	a Aurigae	Capella	5:14:28.2	+45:58:10	0.2	11	Capella								81	a Cas	a Cassiopeiae	Phaedra	0:38:47.5	+56:22:23	2.3	PHAEDRA	ALPHA CASSIOPEIAE
12	a Car	a Carinae	Canopus	6:23:17.1	-52:40:44	-0.9	12	Canopus								82	y UMa	y Ursae Majoris	Phaedra	11:25:15.6	+53:51:41	2.2	PHAEDRA	GAMMA URSAE MAJORIS
13	a Osa	a Canis Majoris	Sirius	6:43:40.6	-16:40:29	-1.6	13	Sirius								83	e Sco	e Scorpis	Antares	15:58:33.3	-22:30:16	2.5	ANTARES	DELTA SCORPII
14	a OMi	a Canis Minoris	Procyon	7:37:43.9	+5:18:11	0.5	14	Procyon								84	e Sco	e Scorpis	Antares	15:58:33.3	-22:30:16	2.5	ANTARES	DELTA SCORPII
15	y Vel	y Velorum	Regor	8:08:36.4	-47:14:51	1.9	15	Regor								85	e Peg	e Pegasi	Antares	01:14:1.3	+15:01:01	2.9	ANTARES	GAMMA PEGASI
16	1 UMa	1 Ursae Majoris	Aldebaran	8:57:09.7	+45:09:38	3.1	16	Aldebaran								86	e Peg	e Pegasi	Antares	01:14:1.3	+15:01:01	2.9	ANTARES	GAMMA PEGASI
17	a Hya	a Hydras	Alphard	9:26:06.8	-8:31:40	2.2	17	Alphard								87	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
18	a Leo	a Leonis	Regulus	10:06:46.5	+12:06:52	1.3	18	Regulus								88	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
19	8 Leo	8 Leonis	Denebola	11:47:33.8	+21:44:23	2.2	19	Denebola								89	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
20	y Cyg	y Cygni	Corvi	12:14:15.6	-17:22:32	2.8	20	Corvi								90	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
21	a Cru	a Crucis	Acrux	12:24:59.4	-62:55:59	1.0	21	Acrux								91	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
22	a Vir	a Virginis	Spica	13:23:36.6	-11:00:19	1.2	22	Spica								92	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
23	n UMa	n Ursae Majoris	Alkaid	13:46:21.6	+49:27:45	1.9	23	Alkaid								93	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
24	a Cen	a Centauri	Menkent	14:04:54.6	-36:13:23	2.3	24	Menkent								94	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
25	a Boo	a Bootis	Arcturus	14:14:17.5	+19:20:16	0.2	25	Arcturus								95	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
26	a Cor	a Coroneae	Alphecca	15:33:22.0	+06:48:53	1.3	26	Alphecca								96	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
27	a Sco	a Scorpis	Antares	16:27:33.9	-26:22:01	1.2	27	Antares								97	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
28	a Tri	a Trianguli Australis	Atria	16:45:28.3	-68:58:31	1.9	28	Atria								98	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
29	a Oph	a Ophiuchi	Rasalhague	17:33:32.4	+12:34:50	2.1	29	Rasalhague								99	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
30	a Igr	a Igneae	Vega	18:35:55.3	+38:45:17	0.1	30	Vega								100	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
31	a Ser	a Serpentis	Nunki	18:53:24.3	-26:20:08	0.9	31	Nunki								101	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
32	a Aql	a Aquilae	Altair	19:49:19.1	+8:47:16	0.9	32	Altair								102	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
33	a Cap	a Capricorni	Peacock	20:19:19.6	-14:52:38	3.2	33	Peacock								103	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
34	a Pav	a Pavonis	Peacock	20:23:17.0	-56:49:58	1.3	34	Peacock								104	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
35	a Cyg	a Cygni	Deneb	20:40:24.4	+45:10:21	1.3	35	Deneb								105	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
36	c Peg	c Pegasi	Enif	21:42:42.7	+9:44:12	2.5	36	Enif								106	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
37	a Psa	a Piscis Austrini	Fomalhaut	22:53:59.7	+29:44:54	1.3	37	Fomalhaut								107	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
38	a Ari	a Arietis	Hamal	2:05:28.6	+23:19:17	2.2	38	Hamal								108	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
39	a For	a Formicis	Formica	3:10:47.7	-29:06:16	3.9	39	Formica								109	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
40	8 Gem	8 Geminorum	Pollux	7:43:29.0	+28:06:00	1.2	40	Pollux								110	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
41	8 Car	8 Carinae	Misak	9:12:52.8	-59:35:37	1.5	41	Misak								111	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
42	a U Ma	a Ursae Majoris	Dubhe	11:01:53.4	+61:54:48	1.9	42	Dubhe								112	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
43	y Cen	y Centauri	Rigel Kent	12:39:51.2	-48:47:43	2.4	43	Rigel Kent								113	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
44	a Cen	a Centauri	Rigel Kent	14:37:32.9	-60:42:46	0.1	44	Rigel Kent								114	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
45	a Ori	a Orionis	Betelgeuse	5:53:32.8	+7:24:10	0 to 1	45	Betelgeuse								115	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
46	8 Cen	8 Centauri	Hadar	14:01:41.4	-60:13:45	0.9	46	Hadar								116	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
47	y Ori	y Orionis	Bellatrix	5:23:31.2	+5:19:26	1.7	47	Bellatrix								117	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
48	8 Tau	8 Tauri	El Nath	5:24:23.6	+28:35:01	1.8	48	El Nath								118	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
49	c Ori	c Orionis	Alnil Lam	5:34:41.4	-1:13:11	1.7	49	Alnil Lam								119	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
50	c Ori	c Orionis	Alnilah	5:39:14.6	-1:17:26	2.0	50	Alnilah								120	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
51	8 Cha	8 Canis Majoris	Marsia	6:21:22.7	+17:56:24	2.0	51	Marsia								121	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
52	y Gem	y Geminorum	Alhena	6:35:58.0	+16:25:35	1.9	52	Alhena								122	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
53	a Cha	a Canis Majoris	Adhara	6:57:16.0	-28:55:49	1.6	53	Adhara								123	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
54	8 Cha	8 Canis Majoris	Al Vazir	7:07:10.3	-26:20:40	1.6	54	Al Vazir								124	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
55	a Gem	a Geminorum	Castor	7:32:41.2	+31:57:19	1.6	55	Castor								125	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
56	c Car	c Carinae	Avior	8:21:54.0	-59:24:45	1.7	56	Avior								126	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
57	8 Vel	8 Velorum	Sargas	8:43:52.5	-34:35:53	2.0	57	Sargas								127	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
58	y Cru	y Crucis	Gacrux	12:29:29.4	-66:56:44	1.6	58	Gacrux								128	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
59	8 Cru	8 Crucis	Becrux	12:45:57.3	-59:31:30	1.5	59	Becrux								129	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
60	c UMa	c Ursae Majoris	Alioth	12:50:42.8	+56:07:20	1.7	60	Alioth								130	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
61	a Sco	a Scorpis	Shaula	17:31:34.1	-37:05:01	1.7	61	Shaula								131	e Sco	e Scorpis	Antares	17:40:24.5	-39:00:59	2.5	ANTARES	GAMMA SCORPII
62	8 Sco	8 Scorpis	Shaula	17:35:09.6	-42:58:50	2.0	62	Shaula								132	e Sco	e Scorpis	Antares	17:40:24.5				

13.12 Vector Comparison Display (MSK 1590)

13.12.1 Function

The Vector Comparison Display shows the local spherical elements, the classical elements and UVW coordinates for a base (first) vector. The differences between the elements of this vector and the second, third, and fourth vectors (if input) will then be computed. From one to four vectors can be specified for comparison. Any available state vector in the evaluation or usable vector tables can be used (see Vector Panel Summary).

The UVW coordinate system, essentially a local vertical, local horizontal coordinate system, is defined as follows:

- X Axis (U): Vector pointing in the direction of the spacecraft's position vector
- Y Axis (V): Vector perpendicular to the x- and z-axes, pointing in the direction of travel for a circular orbit
- Z Axis (W): Vector pointing in the direction normal to the orbit (along the angular momentum vector)

13.12.2 Buttons

TIM: Time of comparison. GMT is assumed if input is positive, GET if negative. Time of V1 vector will be used for comparison if zero is used as the input time.

VEH: Vehicle for the comparison. Only relevant if mission planning mode is active in the MFD and an ephemeris vector was chosen.

V1: Choose ID of first (base) vector from vector panel summary.

V2: Choose ID of second state vector.

V3: Choose ID of third state vector.

V4: Choose ID of fourth state vector.

CLC: Calculate display parameters.

REF: Reference body (Earth or Moon) for the comparison values.

BCK: Back to last menu

13.12.3 Display Parameters

Name	Definition	Unit	Comment
H_a	Keplerian height of apogee (1st column); difference in height of apogee from reference vector (2nd through 4th columns)	n.mi.	Cannot be computed for hyperbolic or parabolic orbits.
H_p	Keplerian height of perigee (1st column); difference in height of perigee from reference vector (2nd through 4th columns)	n.mi.	Cannot be computed for parabolic orbits.
V	Inertial velocity magnitude (1st column); difference in velocity from reference vector (2nd through 4th columns).	ft/sec	
γ	Flightpath angle (1st column); difference in angle from reference vector (2nd through 4th columns).	deg	
ψ	Azimuth (1st column); difference in azimuth from reference vector (2nd through 4th columns).	deg	
ϕ	Latitude (1st column); difference in latitude from reference vector (2nd through 4th columns)	deg	
λ	Longitude (1st column); difference in longitude from reference vector (2nd through 4th columns).	deg	
h	Height above launch pad (Earth) or lunar landing site (1st column); difference in height from reference vector (2nd through 4th columns).	n.mi.	

Name	Definition	Unit	Comment
a	Semimajor axis (1st column); difference in semimajor axis from reference vector (2nd through 4th columns).	n.mi.	Cannot be computed for parabolic orbit.
e	Eccentricity (1st column); difference in eccentricity from reference vector (2nd through 4th columns)	none	
i	Inclination (1st column); difference in inclination from reference vector (2nd through 4th columns)	deg	
ω_p	Argument of perigee (1st column); difference in ω_p from reference vector (2nd through 4th columns)	deg	
Ω	Right ascension of the ascending node (1st column); difference in Ω from reference vector (2nd through 4th columns)	deg	
ν	True anomaly (1st column); difference in true anomaly from reference vector (2nd through 4th columns)	deg	
UVW	UVW position coordinates of vehicle (1st column); difference in UVW coordinates from reference vector in feet (2nd through 4th columns).	n.mi., ft	Differences are computed in feet
$\dot{U}\dot{V}\dot{W}$	velocity components (1st column); difference in UVW velocity from reference vector (2nd through 4th columns).	ft/sec	

13.13 Vector Panel Summary (MSK 1591)

13.13.1 Function

The Vector Panel Summary will display the ephemeris anchor vector identifications, anchor vector times and current GMT. It will also display the vector identification and time for each vector in a Usable Vector Slot and each vector in an Evaluation Vector Slot. GMT of ullage tailoff are displayed for the last executed maneuver for both vehicles. The time tags of all available telemetry vectors will also be displayed. The display is updated every six seconds, so inputs on the page might not cause an instant change to the display.

13.13.2 Display Parameters and Buttons

The left and right half of the display are showing information about state vectors for CSM and LM respectively. The two sides are identical, so the following description applies to both. Each side is divided into several panels: CMC, LGC, AGS, IU, High Speed Radar, Differential Correction (DC) and Last Executed Maneuver. The first four panels show information about state vectors from telemetry and vectors that have been moved to RTCC tables:

UV = Usable Vector Table

EV = Evaluation Vector Table

TH = Telemetry (High Speed)

TL = Telemetry (Low Speed)

The TL vector slot has not been implemented yet. Pressing the TLM button will get data from all computers whose vehicles is set on the configuration page and save them in the high speed telemetry tables. In reality the telemetry vectors get continually updated if telemetry is received, so for a more permanent saving solution the vectors can be moved to the evaluation vector table with the EV buttons. The vector is then moved to that table, assigned a vector ID and is then available for comparison using the vector compare display (MSK 1590). AGS and IU vectors need to have their specific REFSMMATs established for the transfer to the evaluation table to work. The telemetry vectors are not guaranteed to be valid, so the evaluation vectors were checked before being "moved up" to the usable vector table, which can be done with the UV buttons. Once in the usable vector table the vectors are available for ephemeris updates.

The High Speed Radar panel is not currently implemented. It was a vector that is generated automatically at the end of a powered maneuver using radar data and could, if no telemetry vector was available, be a first estimate of the actual cutoff state.

Differential Correction (DC) is the state vector derived from long period ground tracking of the spacecraft during coasting flight. In the RTCC MFD it is simply generated with the DC button and entering the name of the vessel in the simulation. The DC panel will then be updated with a vector ID (APIC/APII for Orbiter API and C/L for the CSM or LM) and a vector time. The HSR and DC vectors are moved directly to the usable vector table.

The last panel shows the last executed maneuver in the mission plan table of the vehicle. The displayed times are ullage on and end of the thrust tailoff period after cutoff.

To cause an ephemeris update the TUP button on each side can be pressed and then a vector from the usable vector table is chosen to do the update.

13.13.3 Errors

If any function of the VPS does not work as expected, an error might have occurred. An associated error message will have been written to the on-line monitor display (MSK 1629). Here follows a list of possible error messages:

<u>Code</u>	<u>Error Description</u>
1	An unrecognizable queue ID was detected.
2	Not used.
3	Not used.
4	The PBI request made was not a valid request (missing information, two sources indicated, etc.).
5	Request to move vector to Usable Slot cannot be honored because there is no vector in the corresponding Evaluation Slot.
6	Request to move vector to ephemeris update cannot be honored because there is no vector in the corresponding Usable Slot.
7	Not used.
8	Test of PBI message header failed; improper message was routed to module.
9	Unidentifiable vector ID in the S84 MED.
10	Vector to be moved to ephemeris update could not be transformed to ECI coordinates.
11	Telemetry vector requested by PBI cannot be furnished this mission.
12	Not used.
13	Not used.
14	The matrix to rotate the telemetry vector requested is not available.
15	Request is for a telemetry vector that is not available in the collection slot for this type vector.
16	Invalid vehicle code from high-speed processor.
17	Invalid vector source code from high-speed processor.
18	Vector with lunar landing site indicator set cannot be routed to CSM ephemeris update.

14 Config Files

The RTCC loads various configuration files to load constants, mission and launch day specific parameters. The files are located at Config\ProjectApollo\RTCC. To load a set of files for the monthly launch window add e.g. "RTCC_MISSIONFILES Apollo 11" to the RTCC section of a prelaunch scenario. This will load the mission constants, TLI and SFP files with the names "Apollo 11 Constants.txt", "Apollo 11 TLI.txt" and "Apollo 11 SPF.txt".

14.1 Star Table

The RTCC star table is stored in the Star Table.txt file. It contains unit vector of all RTCC navigation stars, starting with the star that get used by the Apollo Guidance Computer. The file is equivalent to the RTCC mission table named EZJGSTAR.

14.2 Mission Constants

For each Apollo mission a file with mission constants (e.g. "Apollo 8 Constants.txt") is loaded. These numbers are constant for all possible launch days of a mission. They usually are related to hardware or software which doesn't get changed for a different launch day. Some of these parameters are saved and loaded in scenarios.

Name	Description	Default Value	Unit
AGCEpoch	The epoch of the coordinate system used by the Apollo Guidance Computer and the RTCC	1969 (Apollo 7-10)	Years
TEPHEM0	Starting date of AGC time keeping. Always midnight July 1st of the year preceeding the epoch year. Only needs to be loaded for Skylark and the fixed 1950 coordinate system	40038	Days (MJD)
MCCLEX	Octal LGC address for external DV uplink	3433	None
MCCLRF	Octal LGC address for REFSMMAT uplink (and downlink)	1733	None
MCCCXS	Octal CMC address for desired REFSMMAT uplink	306	None
MCCLXS	Octal LGC address for desired REFSMMAT uplink	3606	None
MCCCRF	Octal CMC address for REFSMMAT uplink (and downlink)	1735	None
MCCLRF	Octal LGC address for REFSMMAT uplink (and downlink)	1733	None
MCLRLS	Octal LGC address for landing site vector uplink (and downlink)	2022	None
MCLTTD	Octal LGC address for the time of lunar landing	2400	None
MCLABT	Octal LGC address for descent abort constants	2545	None
MCCTEP	Octal CMC address for TEPHEM (liftoff time)	1706	None
MCLTEP	Octal LGC address for TEPHEM (liftoff time)	1706	None
PZREAP_RRBIAS	Relative range override for the Return-to-Earth processor	1285.0	Nautical Miles
PZREAP_IRMAX	Maximum return inclination for the Return-to-Earth processor	40.0	Degrees
CSMPadLatitude	Geodetic latitude of the CSM launch pad used for launch and IU REFSMMAT calculation	28.608202 (LC-39A)	Degrees
CSMPadLongitude	Longitude of the CSM launch pad	-80.604133 (LC-39A)	Degrees
LMPadLatitude	Geodetic latitude of the LM launch pad used for IU REFSMMAT calculation	28.531445 (LC-37B)	Degrees
LMPadLongitude	Longitude of the LM launch pad	-80.565077 (LC-37B)	Degrees
MCTVEN	S-IVB LH2 venting scale factor	1.0	None

Name	Description	Default Value	Unit
PDI_v_IGG	PDI ignition algorithm velocity	5545.46	Feet per second
PDI_r_IGXG	PDI ignition algorithm x-axis position	-130519.86	Feet
PDI_r_IGZG	PDI ignition algorithm z-axis position	-1432597.3	Feet
PDI_K_X	PDI ignition algorithm coefficient	0.617631	None
PDI_K_Y	PDI ignition algorithm coefficient	0.755e-6	Feet per Feet ²
PDI_K_V	PDI ignition algorithm coefficient	410.0	Seconds
PDI_RBRFG	PDI braking phase position target	171.835, 0.0, -10678.596	Feet
PDI_VBRFG	PDI braking phase velocity target	-105.876, 0.0, -1.04	Feet per second
PDI_ABRFG	PDI braking phase acceleration target	0.6241, 0.0, -9.1044	Feet per second ²
PDI_JBRFGZ	PDI braking phase jerk	-0.01882677	Feet per second ³
PDI_RARFG	PDI approach phase position target	111.085, 0.0, -26.794	Feet
PDI_VARFG	PDI approach phase velocity target	-4.993, 0.0, 0.248	Feet per second
PDI_AARFG	PDI approach phase acceleration target	-0.2624, 0.0, -0.512	Feet per second ²
PDI_JARFGZ	PDI approach phase jerk	0.00180772	Feet per second ³
MGVDGD	DPS engine gimbal plane in LM coordinates	154	Inches
MGVSGD	SPS engine gimbal plane in CSM coordinates	833.2	Inches
MGVSTD	Distance between SPS and DPS gimbal planes	435.55	Inches
MHVCCG	CSM CG table. Format: entry in table, weight, x, y and z coordinates. Example: 0 26600.0 976.946643 0.614894 3.200135		-, lbs, in, in, in
MHVCCG_N	Number of entries in CSM CG table	20	
MHVLCG	LM CG table. Format: entry in table, weight, x, y and z coordinates. Example: 0 14000.0 222.73 0.0 0.0		-, lbs, in, in, in
MHVLCG_N	Number of entries in LM CG table	40	
MHVACG	LM ascent stage CG table. Format: entry in table, weight, x, y and z coordinates. Example: 0 5000.0 260.9969 0.3187 5.0043		-, lbs, in, in, in
MHVACG_N	Number of entries in LM ascent stage CG table	14	

Name	Description	Default Value	Unit
MGTESE	Number of tesseral and sectorial coefficients in Earth potential	0 (use 4 in Open Orbiter)	
MMTESE	Number of tesseral and sectorial coefficients in Moon potential	0 (use 3 in Open Orbiter)	

14.3 Launch Day Init Parameters

For each launch day of a given mission (e.g. "1968-12-21 Init.txt" for Apollo 8) a few parameters are being loaded as initial values. These parameters are almost all saved and loaded and may be overwritten during a mission, so they are only loaded from file once when the RTCC is first being started.

Name	Description	Default Value	Unit
LSLat	Latitude of the lunar landing site	0.0	Degrees
LSLng	Longitude of the lunar landing site	0.0	Degrees
LSRad	Radius of the lunar landing site	Mean lunar radius	Nautical miles
LDPPDwellOrbits	Number of dwell orbits desired between DOI and PDI used by the Lunar Descent Planning Processor	0	None
LDPPDescentFlightArc	Powered flight arc of descent for the Lunar Descent Planning Processor	15.0	Degrees
LDPPHeightofPDI	Height of PDI used by the Lunar Descent Planning Processor	50000.0	Feet
PZLOIPLN_HP_LLS	Height of perilune at PDI for the LOI Targeting	8.23	Nautical Miles
TLCC_AZ_min	Minimum approach azimuth to the landing site used by the Translunar Midcourse Processor	-110.0	Degrees
TLCC_AZ_max	Maximum approach azimuth to the landing site used by the Translunar Midcourse Processor	-70.0	Degrees
REVS1	Number of orbits between LOI-1 and LOI-2 (or DOI for the later missions) used by the TLMCC and LOI targeting	2.0	None (double)
REVS2	Number of orbits between LOI-2 (or DOI for the later missions) and PDI used by the TLMCC and LOI targeting	11	None (Integer)
LOPC_M	Number of revolutions from first pass over lunar landing site to lunar orbit plane change maneuver. Used by LOPC routine of the TLMCC processor	3	None
LOPC_N	Number of revolutions from lunar orbit plane change maneuver to second pass over lunar landing site. Used by LOPC routine of the TLMCC processor	8	None

Name	Description	Default Value	Unit
LOI_psi_DS	Desired approach azimuth to the lunar landing site. Used by the LOI targeting	270.0	Degrees
eta_1	True anomaly on LPO-1 (orbit after LOI-1) for transferring from the hyperbola to LPO-1. Used by TLMCC and LOI targeting	0.0	Degrees
H.P_LPO1	Perilune height on LPO-1 (orbit after LOI-1). Used by TLMCC and LOI targeting	60.0	Nautical miles
PZLTRT_DT_Ins_TPI	Time between insertion and TPI for the short rendezvous profile in lunar orbit	40.0	Minutes
SITEROT	Angle of perilune from the lunar landing site (negative if the site is post-perilune). Used by the TLMCC and LOI targeting	-15.0	Degrees

14.4 Skeleton Flight Plan Table

The skeleton flight plan table is a table of initial guesses for the translunar midcourse correction processor. The files are specific to the monthly launch window and contain up to 10 days of data. The file names are usually something like "Apollo 11 SFP.txt". The format of the file follows the MSC card format from the MSC internal note RTCC REQUIREMENTS FOR APOLLO 11 (MISSION G) PREFLIGHT INFORMATION (69-FM-171), which can be found [here](#). PDF page 13 shows the general format of one data set. One line in the config file is equal to one punch card. Some adaptations have been made for the additional parameters needed on hybrid missions. The complete format for one data set is:

Day (1 to 366)	Opportunity (1 or 2)	Launch azimuth (deg)	Launch window (A or P)	Card ID
Latitude of TLI pericyynthion (deg)	Longitude of TLI pericyynthion (deg)	Height of TLI pericyynthion (NM)	GET of TLI (hours)	Card ID
Latitude of LOI pericyynthion (deg)	Longitude of LOI pericyynthion (deg)	Height of LOI pericyynthion (NM)		Card ID
dpsi of LOI (deg)	gamma of LOI (deg)	Time in lunar orbit (hrs)	Time from LOI to landing (hrs)	Card ID
Approach azimuth (deg)	Latitude of the landing site (deg)	Longitude of the landing site (deg)	Radius of the landing site (NM)	Card ID
dpsi of TEI (deg)	DV of TEI (ft/s)	time from TEI to EI (hrs)	Inclination of free return (deg)	Card ID
GMT of TLI pericyynthion (hrs)	GMT of LOI pericyynthion (hrs)	GMT of node (hrs)		Card ID
Latitude of node (deg)	Longitude of node (deg)	Height of node (NM)		Card ID

14.5 TLI Targeting Parameters

The files containing the TLI targeting parameters follow the launch day specific format "1968-12-21 TLI.txt". They contain the numbers needed to calculate the time of restart preparations and the TLI maneuver in the RTCC. The format of the file follows the MSC card format from the MSC internal note RTCC REQUIREMENTS FOR APOLLO 11 (MISSION G) PREFLIGHT INFORMATION (69-FM-171), which can be found [here](#). The PDF pages 32 to 40 show the format. One line in the config file is equal to one punch card.

14.6 Erasable Memory Programs

Erasable Memory Program (EMP) files are loaded in the EMPs folder, which is under Config/ProjectApollo/RTCC. They contain uplink data for the CMC and LGC. The format of the file is as follows:

- The first line contains a description of the EMP which is shown on the uplink MFD page.
- The second line contains the AGC rope name (e.g. Luminary 210) for which the EMP is valid.
- The actual uplink loads follow in the subsequent lines. Each load has two lines in the file and the number of possible loads per file is not limited. There are two types of EMP uplinks. Contiguous block update (Verb 71) updates from 1 to 18 consecutive erasable memory registers in the same erasable memory bank. The scatter update (Verb 72) provides the capability to update from 1 to 9 non-consecutive erasable memory registers in the same or different erasable memory banks. The first of the two lines in each load contains either 71 and the starting address for Verb 71 or 72 and no additional value for Verb 72. The second line then contains all the uplink data, for Verb 72 it is alternating address and value. All data is in octal.

On the next page follows a list of EMPs that come with NASSP:

EMP Filename	Description	Rope
P99 L1D	Guided RCS Translation Maneuver	Luminary 178
EMP 99	Guided RCS Translation Maneuver	Luminary 210
EMP 100A	Backup for failed DSKY key using engine gimbal switch	Luminary 210
EMP 100B	Backup for failed DSKY key using mode sel switch	Luminary 210
EMP 103A	Descent with failed CDUs	Luminary 210
EMP 103B	EMP for failed CDUX	Luminary 210
EMP 108	Zero a runaway IMU CDU and prevent coarse align	Luminary 210
EMP 500	Landmark Tracking With Datalink Failure Or Unusable Optics	Artemis 72
EMP 501	Landmark Tracking With Failed Mark/Mark Rej Button	Artemis 72
EMP 512	P40/41/47 Termination During Average G When EMP 509 is Operating	Artemis 72
EMP 514	Shortened P23	Artemis 72
EMP 515	Manual Range Input	Artemis 72
EMP 517	Convert Optics Shaft And Trunnion Angles To Body Angles	Artemis 72
EMP 523	Monitor Jet-On Failure	Artemis 72
EMP 525A	Optics Switch Monitor	Artemis 72
EMP 526 (1 Deg)	IMU CDU Transient Monitor (1 Degree Check)	Artemis 72
EMP 526 (2 Deg)	IMU CDU Transient Monitor (2 Degree Check)	Artemis 72
EMP 526 (3 Deg)	IMU CDU Transient Monitor (3 Degree Check)	Artemis 72
EMP 526 (4 Deg)	IMU CDU Transient Monitor (4 Degree Check)	Artemis 72
EMP 527 (CDUX)	Monitor Single IMU CDU (CDUX)	Artemis 72
EMP 527 (CDUY)	Monitor Single IMU CDU (CDUY)	Artemis 72
EMP 527 (CDUZ)	Monitor Single IMU CDU (CDUZ)	Artemis 72
EMP 528	Monitor Jet-On Failure and Do EMP 526	Artemis 72
EMP SL-15	Manual Range Input	Skylark 48
EMP SL-23	Monitor Jet-On Failure	Skylark 48
EMP SL-26 1 Deg	IMU CDU Transient Monitor (1 Degree Check)	Skylark 48
EMP SL-26 2 Deg	IMU CDU Transient Monitor (2 Degree Check)	Skylark 48
EMP SL-26 3 Deg	IMU CDU Transient Monitor (3 Degree Check)	Skylark 48
EMP SL-26 4 Deg	IMU CDU Transient Monitor (4 Degree Check)	Skylark 48
EMP SL-27 CDUX	Monitor Single IMU CDU (CDUX)	Skylark 48
EMP SL-27 CDUY	Monitor Single IMU CDU (CDUY)	Skylark 48
EMP SL-27 CDUZ	Monitor Single IMU CDU (CDUZ)	Skylark 48
EMP SL-28	Monitor Jet-On Failure and Do EMP SL-26	Skylark 48
EMP SL-50	Convert Gyro Torquing Angles to CDU Angles for Failed CDU	Skylark 48
EMP SL-52A	Checksum for VAC area 4 for EMP SL-23	Skylark 48
EMP SL-52B	Checksum for VAC area 4 for EMP SL-28	Skylark 48
EMP SL-55	IMU Slip Ring Program	Skylark 48
EMP ASTP-75	Compute S-Band Antenna Angles	Skylark 48
EMP ASTP-77	Raster Scan	Skylark 48